



ACQLERATE

MODULE 11

# Beyond Award: Sustainment, Test, Software, and Closeout

*What happens after the signature: operating and support cost, test and evaluation, the software acquisition pathway and agile contracting, contract closeout, terminations for convenience and default, incumbent-to-successor transitions, options and bridges, and the program reviews and baselines that keep a program honest.*

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8 Lessons · ~192 Minutes · Lesson Book Edition

## Course Lessons

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|    |                                                                               |        |
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| 01 | Sustainment and O&S Cost: Where Most of the Money Goes                        | 25 min |
| 02 | Test and Evaluation: How the Government Finds Out Whether It Works            | 25 min |
| 03 | The Software Acquisition Pathway and Agile Under a Government Contract        | 26 min |
| 04 | Contract Closeout: The Work After the Work                                    | 22 min |
| 05 | Terminations: Convenience, Default, and What Each Costs                       | 25 min |
| 06 | Contract Transition: Taking Over and Handing Off Without Dropping the Mission | 24 min |
| 07 | Options, Extensions, and Bridges: Keeping a Contract Alive Legally            | 22 min |
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## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 1

# Sustainment and O&S Cost: Where Most of the Money Goes

25 min

*Development and production get the attention; operating and support gets the money. What O&S cost is and why it dominates life-cycle cost, the sustainment decisions made during acquisition that lock it in, product support strategies from organic depot to performance-based logistics, the statutory rules on depot work, and what contractors and government PMs each control.*

## The Part of the Program Nobody Gets Promoted For

A weapon system's development might take eight years and its production ten. It will then be operated and supported for thirty or forty. Over that life, the cost of operating and supporting it, fuel, spares, repair, personnel, training, software updates, facilities, typically exceeds the cost of developing and producing it, often by a wide margin. And most of that cost is determined by decisions made during development: the design's reliability, the maintenance concept, the data rights, the supply chain, the support strategy. The Foundations lessons cover the lifecycle phases; this lesson is about the phase that costs the most and gets the least attention, and about the decisions in the early phases that decide it.

### IN PLAIN TERMS

Think about buying a car. The sticker price is what everyone negotiates, but over fifteen years you also pay for gas, insurance, tires, oil changes and repairs. A cheaper car that breaks often, or needs parts only one dealer sells, can cost far more to own than it did to buy. Operating and support (O&S) cost works the same way: the design choices made before purchase decide most of what the system costs over its 30 to 40 years in service.

## Where Life-Cycle Cost Goes (Typical Major System)

### 10 to 15%

Development

RDT&amp;E; where the attention is

### 25 to 35%

Production

Procurement; where the politics are

### 50 to 70%

Operating and support

O&amp;M and MILPERS; where the money is

### 30 to 40 years

Service life

The period O&amp;S runs

## What O&S Cost Is Made Of

DoD's cost estimating guidance breaks O&S into standard elements: unit-level manpower (the operators and maintainers), unit operations (fuel, consumables, training munitions), maintenance (organizational,

intermediate, and depot; the parts and the labor), sustaining support (engineering, software maintenance, technical data updates, simulators), continuing system improvements (modifications and upgrades), and indirect support (base operations and personnel support attributed to the system). For a services-heavy system like a software platform, sustaining support and continuing improvements dominate. For an aircraft, fuel, maintenance, and manpower do. The estimate is built by element, by year, over the service life, and it is the largest number in the program's life-cycle cost estimate.

## Decisions Made Early That Set O&S Cost

| DECISION                                              | WHEN IT IS MADE                         | O&S CONSEQUENCE                                                                          |
|-------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| Reliability and maintainability requirements          | Requirements and design                 | Every failure avoided is a repair, a spare, and downtime avoided for forty years         |
| Maintenance concept (what is repaired where, by whom) | Design and LCSP                         | Determines depot workload, contractor role, and the training and facilities needed       |
| Technical data and software rights                    | Contract award (the Data Rights lesson) | Whether sustainment can be competed or is sole source to the OEM for life                |
| Commonality and open architecture                     | Design                                  | Whether components can be competed and replaced, or must come from the original supplier |
| Supply chain and DMSMS planning                       | Production                              | How often parts go obsolete and what it costs to redesign around them                    |
| Support strategy (organic, contractor, PBL)           | LCSP at Milestone B and C               | Cost structure, incentives, and statutory compliance for the life of the system          |

### EXAMPLE (ILLUSTRATIVE)

Program X will field 200 vehicles for 30 years. During design, the program office compares two options. Design A is expected to need about 4 repairs per vehicle per year. Design B, with better reliability built in, needs about 3. Assume each repair costs \$5,000 in parts and labor (a hypothetical figure). One repair avoided per vehicle per year adds up:  $200 \text{ vehicles} \times 1 \text{ repair} \times 30 \text{ years} \times \$5,000 = \$30,000,000$ . That is before counting fewer spares, less downtime and fewer maintainers. If Design B costs a few million dollars more during development, it still saves money over the system's life. This is why reliability requirements are an O&S decision, not just an engineering one.

## Product Support Strategies: From Organic to PBL

The government has a spectrum of choices for who supports a fielded system. At one end, organic support: government depots repair it, government supply systems stock it, government personnel maintain it, with contractors providing parts and engineering under transactional contracts. At the other, contractor logistics

support: a contractor provides most of the maintenance and supply under a long-term contract. Between them, mixed strategies assign elements to whichever source is best. Performance-based logistics is an incentive structure that can apply anywhere on the spectrum: instead of buying parts and repair hours, the government buys an outcome (an availability rate, a response time, a cost per flying hour) and pays the provider for achieving it. The provider then has the incentive to improve reliability and reduce repairs, because it keeps the savings. PBL works when the outcome is measurable, the provider controls the drivers, and the contract is long enough for reliability investments to pay off.

#### THE DATA RIGHTS BILL ARRIVES FIFTEEN YEARS LATER

Sustainment is where the data rights decision made fifteen years earlier arrives with an invoice. A program that delivered without securing interface data and depot-level technical data cannot compete a single spare or a single repair, and the OEM knows it. The largest sustainment cost driver is not the design. It is whether anyone other than the original contractor can touch the system.

### The Statutory Rules That Shape the Strategy

- **Core logistics capability (10 U.S.C. 2464)** DoD must maintain a government depot capability for systems essential to readiness, identified within a set period after fielding. The support strategy has to plan for organic depot work on covered systems, which constrains contractor logistics support.
- **The 50/50 rule (10 U.S.C. 2466)** No more than half of each military department's depot maintenance funding in a year may go to contractors. Program strategies are shaped by where the department stands against the limit.
- **Depot source of repair decisions** Formal determinations of where depot-level repair will be performed, made during acquisition and reviewed as systems mature.
- **PSM and LCSP requirements** Major programs must have a product support manager and a life cycle sustainment plan, reviewed at milestones and independently assessed on the largest programs.
- **Sustainment reviews** Statute requires periodic reviews of fielded major systems comparing actual O&S cost and availability to the estimates and the requirements, with corrective actions.

One sentence to teach your team: **operating and support is the largest share of life-cycle cost, it is set by reliability, maintenance concept, data rights, and support strategy decisions made during development, and the government has statutory limits on how much of it can be contracted.**

## INTERMEDIATE

## The Twelve Integrated Product Support Elements

The LCSP is organized around twelve elements, and a sustainment strategy assigns each to a source and a plan: product support management; design interface (the reliability and maintainability built in); sustaining engineering; supply support (spares and provisioning); maintenance planning and management; packaging, handling, storage, and transportation; technical data; support equipment; training and training support; IT systems continuous support; facilities and infrastructure; and manpower and personnel. A PM reading an LCSP checks that each element has an owner, a funding line, and a metric. The elements most often underfunded are technical data (because it was not bought), sustaining engineering (because it is invisible until something fails), and software support (because the program assumed the software was finished).

## PBL Done Right and Done Wrong

A well-designed PBL arrangement has a measurable outcome the provider can influence (availability of a subsystem, not of the whole fleet if the provider only supports one part), a baseline established from real data, a contract long enough to reward reliability investment (five years with options is common), incentives that share savings, and government insight into the provider's cost so that the price can be renegotiated as reliability improves. Done wrong, PBL pays a fixed price for an availability target the provider was already meeting, with no cost insight, so the government never sees the savings; or it sets an outcome the provider does not control, so the incentive is meaningless. The Contract Types lesson and the Award Fee lesson cover incentive mechanics; sustainment is where they run longest.

## Sustainment Metrics

| METRIC                                  | WHAT IT MEASURES                                                | WHO USES IT                                 |
|-----------------------------------------|-----------------------------------------------------------------|---------------------------------------------|
| Materiel availability                   | Share of the total inventory operationally capable              | Requirements documents; sustainment reviews |
| Operational availability                | Share of time an assigned system is mission-capable             | Operators; PBL contracts                    |
| Materiel reliability                    | Mean time between failures                                      | Design interface; DMSMS and spares planning |
| Mean down time                          | Average time a system is not capable, including logistics delay | Supply and maintenance planning             |
| O&S cost (per unit, per hour, per year) | Actual cost against the estimate                                | Sustainment reviews; should-cost            |
| Logistics response time                 | Time from demand to delivery of a part or repair                | PBL contracts; supply chain management      |

## The Contractor's Sustainment Business

For a contractor, sustainment is often the best business on the program: long, predictable, less competed, and with margins that reflect the OEM's advantages. It is also where the Business module's lessons on backlog, cash, and valuation land: a sustainment contract with years of options is durable backlog; a PBL contract with reliability incentives can be highly profitable if the contractor invests in the design; and an acquirer values sustainment revenue highly because it is sticky. The government knows all of this, which is why it pushes for data rights, open architectures, and competition in sustainment, and why a contractor's sustainment position is worth protecting through performance rather than through exclusivity that the government will eventually break.

### ADVANCED

## Sustainment Cost Growth: Why It Happens

O&S cost estimates made at Milestone B are routinely exceeded, for identifiable reasons. Reliability falls short of the requirement, so failures and repairs exceed the estimate. Operating tempo differs from the assumption. Parts go obsolete faster than planned and redesigns cost more than spares. Software support was estimated as a fraction of development and turns out to be a permanent engineering team. Data rights were not secured and the OEM's sole-source prices rise. Manpower assumptions relied on automation that did not materialize. And the estimate itself was optimistic because sustainment cost was not what the milestone decision was about. Senior PMs on both sides track actual O&S cost against the estimate from the first year of fielding, because the trend in year two predicts the sustainment review in year eight.

## Competing Sustainment: What It Takes

Government PMs who want competition in sustainment need four things in place before the system is fielded: technical data with rights sufficient for a third party to repair and manufacture spares (at least form, fit, and function plus depot-level repair data, ideally more); an architecture with published interfaces so components can be replaced without the OEM; a qualified second source for high-value spares, established during production when the volume justifies it; and organic depot capability where the core rules require it. Each has a cost during development that the program will be tempted to cut. Each saves multiples of that cost during sustainment. The acquisition strategy is where the decision is made, and the Data Rights and Acquisition Planning lessons explain how.

## Sustainment From the Contractor Side: Building the Position

How a contractor earns a durable sustainment role without depending on lock-in.

**Design for the metric** RELIABILITY

Reliability and maintainability investments during development pay back through PBL incentives and fewer warranty claims.

The contractor that improves availability keeps the savings under PBL and the contract at recompet.

**Own the supply chain knowledge** DMSMS

Tracking obsolescence and proposing solutions before parts disappear is a service the government values and few competitors can offer.

A DMSMS management program is a sustainment discriminator.

**Support the organic depot** PARTNERSHIP

Public-private partnerships with government depots satisfy core and 50/50 rules while keeping the contractor in the work.

Fighting the depot loses; partnering with it lasts.

**Price for the long game** COST

A sustainment position priced to win the option years and the follow-on, not to maximize year one.

The government tracks O&S cost against the estimate. Surprises invite competition.

**Make data rights a negotiation, not a wall** RIGHTS

Offering a specifically negotiated license at a price preserves value and goodwill; refusing everything invites the government to design around you.

The Data Rights lesson covers the mechanics. The strategy is to be paid for the data rather than to lose the program over it.

**WHAT SENIOR PMS DO DIFFERENTLY**

Government: they treat the LCSP as a real plan with owners, funding, and metrics; they buy the data and the interfaces during development; they design PBL around outcomes the provider controls, with cost insight; and they track O&S actuals from the first year of fielding. Contractor: they design for the sustainment metric, manage obsolescence as a service, partner with organic depots, and price sustainment for the long game rather than the first year.

**BUILD ON THIS**

- **Data Rights and IP** The decision that determines whether sustainment can be competed
- **The Acquisition Lifecycle** Where operations and support sits in the lifecycle
- **Backlog, Book-to-Bill, and Funded vs. Unfunded** Why sustainment backlog is valued highly

## Key Terms

### Operating and Support (O&S) Cost

The cost of operating, maintaining, and supporting a system over its service life: fuel, spares, repair, personnel, training, facilities, software support. Typically the largest share of total life-cycle cost, often well over half.

### Life-Cycle Cost

All costs from research through disposal: development, production, O&S, and disposal. The number acquisition decisions are supposed to optimize, and the one they rarely do.

### Life Cycle Sustainment Plan (LCSP)

The program document describing how the system will be supported: the product support strategy, the metrics, the funding, the organic and contractor roles. Required at milestones and updated through the life cycle.

### Product Support Manager (PSM)

The government official on a major program responsible for the sustainment strategy, statutory since 2009. The PM's counterpart for everything after fielding.

### Product Support Strategy

The mix of organic (government depot and personnel) and contractor support chosen for a system, across the twelve integrated product support elements (supply, maintenance, training, technical data, and so on).

### Performance-Based Logistics (PBL)

A support arrangement that buys outcomes (availability, readiness, response time) rather than transactions (parts and hours), with the provider deciding how to deliver them and incentives tied to the outcome.

### Core Logistics Capability

The statutory requirement (10 U.S.C. 2464) that DoD maintain a government-owned, government-operated capability to maintain systems essential to readiness, which limits how much depot maintenance can be contracted.

### 50/50 Rule

The statutory limit (10 U.S.C. 2466) that no more than 50% of a military department's depot maintenance funding may be spent on contractor performance in a fiscal year.

### Diminishing Manufacturing Sources and Material Shortages (DMSMS)

The loss of suppliers or materials for parts still needed to support a fielded system. A recurring sustainment cost driver, especially for electronics.

### Availability

The share of time a system is capable of performing its mission (materiel availability, operational availability). The sustainment metric that matters most and the one PBL arrangements are usually built on.

## Knowledge Check

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What share of a typical major system's life-cycle cost is operating and support?

- A. About 10%
- B. About 25%
- ✓ C. Often 50% to 70%
- D. Less than development

### WHY

O&S over a 30 to 40 year service life typically exceeds development and production combined.

Which early decision has the largest effect on whether sustainment can be competed?

- A. The color of the paint
- ✓ B. Technical data and software rights secured at contract award
- C. The number of test articles
- D. The award fee percentage

### WHY

Without data rights and interfaces, sustainment is sole source to the OEM for the life of the system.

What does performance-based logistics buy?

- A. Parts and repair hours
- ✓ B. Outcomes such as availability or response time, with the provider deciding how to deliver them and incentives tied to the result
- C. Only organic depot work
- D. Development services

### WHY

PBL aligns the provider's incentive with reliability, because the provider keeps the savings from fewer repairs.

**What is the 50/50 rule?**

A. Half of development must be competed

✓ **B. No more than 50% of a military department's depot maintenance funding may go to contractor performance in a fiscal year**

C. Half of spares must be domestic

D. PBL contracts must be 50% incentive

**WHY**

10 U.S.C. 2466 constrains how much depot work can be contracted, alongside the core logistics requirement.

**What is the Life Cycle Sustainment Plan?**

A. The contractor's warranty

✓ **B. The program document describing the product support strategy, metrics, funding, and organic and contractor roles, required at milestones**

C. The IGCE for O&S

D. A DAU course

**WHY**

The LCSP is organized around the twelve integrated product support elements, each with an owner, funding, and a metric.

**Which IPS elements are most often underfunded?**

A. Facilities and packaging

✓ **B. Technical data, sustaining engineering, and software support**

C. Training

D. Manpower

**WHY**

Data was not bought, engineering is invisible until failure, and software was assumed finished.

**When does PBL work well?**

A. Always

✓ **B. When the outcome is measurable, the provider controls the drivers, the contract is long enough to reward reliability investment, and the government has cost insight**

C. Only for aircraft

D. When the price is fixed with no cost insight

**WHY**

PBL fails when the target was already met, the provider cannot influence it, or the government never sees the savings.

**What is DMSMS?**

A. A depot maintenance metric

✓ **B. The loss of suppliers or materials for parts still needed to support a fielded system**

C. A software pathway

D. A milestone review

**WHY**

Obsolescence is a recurring sustainment cost driver, especially for electronics, and managing it is a contractor discriminator.

**Why do O&S cost estimates made at Milestone B get exceeded?**

A. They never are

✓ **B. Reliability shortfalls, operating tempo, obsolescence, underestimated software support, unsecured data rights, and optimistic assumptions**

C. Inflation only

D. Contractor fraud

**WHY**

Each cause is identifiable, and the trend in early fielding years predicts the sustainment review.

**Match the decision to its O&S consequence:****CORRECT MATCHES**

|                          |                                                    |
|--------------------------|----------------------------------------------------|
| Reliability requirements | Repairs, spares, and downtime for the service life |
| Maintenance concept      | Depot workload and contractor role                 |
| Data rights at award     | Whether sustainment can be competed                |
| Open architecture        | Whether components can be replaced without the OEM |

**WHY**

Reliability sets repairs and spares; maintenance concept sets depot and contractor roles; data rights set competition; open architecture sets component replaceability.

**Order what a government PM needs in place to compete sustainment:****CORRECT ORDER**

1. Secure technical data rights at contract award
2. Build an architecture with published interfaces
3. Qualify a second source for high-value spares during production
4. Stand up organic depot capability where core rules require

**WHY**

Data rights first (at award), then open interfaces in design, then a second source during production, then organic depot capability where required.

**How does a contractor build a durable sustainment position without relying on lock-in?**

A. By refusing all data rights

**✓ B. By designing for reliability, managing obsolescence as a service, partnering with organic depots, pricing for the long game, and negotiating data licenses rather than walls**

C. By maximizing year-one price

D. By avoiding PBL

**WHY**

Exclusivity the government will eventually break is worth less than a position earned through performance.

## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 2

# Test and Evaluation: How the Government Finds Out Whether It Works

25 min

*Developmental and operational testing, the technical reviews that gate them, the test and evaluation master plan, the independent evaluators, and what a test event means for cost, schedule, and the contract. Written for the PM who has to plan around testing and the contractor who has to pass it.*

## Testing Is Where the Program Meets Reality

Every requirement the program wrote, every design decision the contractor made, and every schedule the PM briefed is, eventually, tested. A system either meets its specifications or it does not; it either works in the hands of operators or it does not. Test and evaluation is the discipline that finds out, and it is structured so that the finding is credible: the developer tests first, then an independent agency tests under realistic conditions, and an independent office reports the result to Congress. For a PM, testing is the schedule's hardest constraint and the source of most late surprises. For a contractor, it is the event that determines whether the design was right and whether the next contract comes. This lesson covers the structure, the reviews that gate it, and what a test event means for the money.

### IN PLAIN TERMS

Think of learning to drive. First you practice with an instructor in an empty parking lot, checking that you can brake, signal and park. Then a state examiner, who has no stake in whether you pass, rides with you in real traffic. Developmental test and evaluation (DT&E) is the parking lot: does the system meet its specifications under controlled conditions? Operational test and evaluation (OT&E) is the road test: an independent agency watches typical users run it in realistic conditions.

## DT&E vs. OT&E

|                 | DEVELOPMENTAL TESTING                                                               | OPERATIONAL TESTING                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Question        | Does the system meet its technical specifications?                                  | Is it effective, suitable, and survivable in realistic use by typical operators?                        |
| Who runs it     | The program and the contractor, with government DT organizations                    | An independent operational test agency (each service has one); DOT&E oversees major programs            |
| Conditions      | Controlled; instrumented; often contractor facilities and government ranges         | Realistic operational environment; production-representative articles; typical users; realistic threats |
| Output          | Verification of specifications; deficiency reports; readiness for OT                | Evaluation of effectiveness and suitability; the report that supports the full-rate production decision |
| When            | Throughout development, through the engineering and manufacturing development phase | Before the full-rate production decision (initial OT&E); after fielding (follow-on OT&E)                |
| Contractor role | Central: builds, tests, fixes, supports                                             | Limited: provides articles and support; may not operate or influence the test                           |

## The Technical Reviews That Gate Testing

Testing does not begin until the design has passed a sequence of reviews, each with entry and exit criteria. The system requirements review confirms the requirements are understood and allocated. The preliminary design review confirms the design approach will meet them and that risks are identified. The critical design review confirms the detailed design is complete and ready to build. The test readiness review confirms that the test articles, procedures, instrumentation, safety, and personnel are ready for a specific event. Each review is a gate: passing it releases the next phase of work and, often, the next contract action. Failing it stops the clock. A PM who schedules testing without scheduling the reviews and their preparation has scheduled a slip.

## The Test and Evaluation Master Plan

The TEMP integrates everything into one document, approved at milestones. Its parts:

- **Evaluation framework** How each requirement (KPPs, key system attributes, critical technical parameters) will be evaluated: which test, which measure, which criterion.
- **Test program** The events, in sequence: contractor testing, government DT, integrated events, OT, live fire where required, with what articles and where.
- **Resources** Test articles, ranges, instrumentation, threat representations, personnel, and their cost and schedule. The resource section is where test drives the budget.
- **Schedule** Events tied to the program schedule and the milestone decisions they support.

- **Roles** Who plans, executes, and evaluates each event, and who approves the plans.

## What a Test Event Costs and Why It Slips

Test events are expensive: articles that must be built and instrumented, ranges that must be scheduled months ahead, threat systems and targets, data collection and analysis, and the people. A major test campaign can be a meaningful share of a development contract's cost. And they slip, for predictable reasons: the article is not ready (the TRR fails or is skipped), the range is not available, the instrumentation does not work, the weather does not cooperate, or an earlier event found a deficiency that has to be fixed first. Each slip cascades: the next event, the review that depends on it, the milestone that depends on the review. A PM budgets schedule margin around test events and treats a passed TRR as the real entry criterion.

### PROGRAMS THAT SKIP THE TRR TEST TWICE

The most expensive test is the one that runs before the system is ready. A failed event consumes the range time, the articles, and the budget, and produces a deficiency report that everyone already knew was coming. The TRR exists to prevent that. Programs that treat the TRR as a formality, because the schedule says the test is next week, are the programs that test twice.

## Deficiencies: The Output That Drives the Contract

Testing produces deficiency reports: findings that the system did not perform as required, categorized by severity (whether the deficiency prevents mission accomplishment, degrades it, or is minor). Each gets a disposition: fix before the next event, defer to a later increment, or accept with a requirements change. The dispositions are contract actions. A fix is either within the contractor's existing obligation (the specification was not met) or a change (the requirement was ambiguous or the government changed it), and the two are argued about. The Modifications lesson covers changes; here the point is that the deficiency report is where the argument starts, and a clear specification and a clear test criterion are what settle it.

### EXAMPLE (ILLUSTRATIVE)

Program X is buying a radio. The specification says it must transmit at least 50 kilometers in stated conditions. In developmental testing it reaches only 42 kilometers. A deficiency report is written, and because the specification was clear and was not met, the fix is within the contractor's existing obligation. The contractor fixes it at its own cost before the next event. A second deficiency report says the radio is hard to operate while wearing gloves. The specification never mentioned gloves. Fixing this is a change, because the government is adding a requirement, so it becomes a contract modification with a price. The two reports look alike, but the wording of the specification decided who pays.

One sentence to teach your team: **developmental testing verifies the specification and operational testing evaluates the system in real use by an independent agency, both are gated by technical reviews with entry criteria, and the deficiency reports they produce are where the contract's obligations get tested too.**

#### INTERMEDIATE

### Independence: Why OT&E Is Run by Someone Else

Operational testing is independent by law. Each military department has an operational test agency that reports outside the acquisition chain; DOT&E at OSD oversees testing on major programs, approves their test plans, and sends its own assessment to Congress and the Secretary. The contractor does not run the test and may not operate the system during it. The program office does not control the evaluation. This independence exists because programs under schedule pressure have, historically, declared systems ready that were not, and because the users who will depend on the system deserve an evaluation from someone who does not need the program to succeed. For the PM, it means the OT report is not negotiable; the way to get a good one is to be ready.

### Integrated Testing and the Push for Efficiency

Testing is expensive enough that DoD has pushed for integrated testing: designing DT and OT so that events share articles, ranges, and data, so that DT data can support OT conclusions where conditions are realistic enough, and so that the total test program is shorter. The independent agencies participate in DT planning and observe DT events for this reason. The limit is credibility: OT conclusions require operational realism, typical operators, and independence, and DT data collected under contractor control does not always qualify. The TEMP is where the integration is designed, and the program that involves the operational testers early gets more credit for its DT data than one that presents it afterward.

## The Reviews, Their Questions, and What Failing Them Costs

| REVIEW           | QUESTION                                                                       | TYPICAL FAILURE                                           | COST OF FAILING                                                     |
|------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------|
| SRR              | Are the requirements complete, understood, and allocated?                      | Requirements still changing; interfaces undefined         | Design proceeds on shifting requirements; rework at PDR             |
| PDR              | Will the design approach meet the requirements with acceptable risk?           | Immature technology; unresolved trades                    | Detailed design proceeds on an approach that changes; cost growth   |
| CDR              | Is the detailed design complete and ready to build?                            | Drawings incomplete; analyses missing; interfaces open    | Fabrication starts on an incomplete design; test articles are wrong |
| TRR              | Are the article, procedures, instrumentation, and people ready for this event? | Article not configured; procedures untested; range issues | A failed or invalid event; range time lost; schedule cascade        |
| Post-test review | What did the event show and what are the dispositions?                         | Deficiencies without owners; data not analyzed            | Fixes not made before the next event; repeat findings               |

## Software and Testing

For software-intensive systems, testing is continuous rather than event-based: automated testing in the development pipeline, security testing for the authorization to operate, and user evaluation of increments as they are delivered. The Software Pathway lesson covers the acquisition side. For testing, the change is that the TEMP describes a continuous evaluation approach rather than a sequence of discrete events, and that operational evaluation happens on fielded increments with real users. The independence principle still applies; the independent agencies have adapted their methods to evaluate software delivered in sprints rather than in a single OT campaign.

### ADVANCED

## The Contractor's Test Strategy

A contractor on a development program plans its testing to pass the government's, and the discipline is to test earlier and harder than the government will. Component and subsystem testing before integration, so integration problems are design problems already solved. Full qualification testing against the specification before government DT, so the government sees a system that has already met its numbers. Reliability growth testing with enough hours to demonstrate the requirement with statistical confidence, because reliability is where operational suitability findings come from and it cannot be fixed in the last month. And realistic use by operator-representative users before OT, because the deficiencies operators find are not the

ones engineers find. A contractor that arrives at government testing having already found and fixed its problems passes; one that uses government testing to find them does not.

## Test Failures and the Program

A failed KPP in operational testing is a program event: it can stop the full-rate production decision, trigger a requirements review, require a redesign, and become a line in DOT&E's annual report to Congress. Senior PMs manage this by testing to the KPP thresholds early and often in DT, so that the OT result is not a surprise; by preparing the requirements community for the possibility that a threshold needs adjustment, with the analysis to support it; and by having the contract mechanism in place (incentives, warranty, correction of deficiencies clauses) for the fixes that will be needed. The alternative is discovering in OT what should have been known at CDR, with the whole program waiting.

## Test Planning Failures

From lessons-learned reports across programs.

### Test articles that are not production-representative ARTICLES

OT on prototypes or early builds that differ from what will be fielded.

The evaluation does not apply to the fielded system, and OT must be repeated.

### No reliability growth plan RELIABILITY

Reliability demonstrated only at the end, with too few hours.

Suitability failure in OT. Reliability requires a growth curve planned from the start.

### Range and resource assumptions RESOURCES

Range time, threat representations, and instrumentation assumed available on the program's schedule.

Ranges are scheduled years out. The TEMP resource section is where the reality goes.

### Skipped or rubber-stamped TRRs READINESS

Events started to hold the schedule with the article not ready.

The event fails or is invalid, and the schedule it was protecting slips further.

### Late involvement of the operational testers INDEPENDENCE

OT agency sees the system for the first time at the OT readiness review.

DT data gets no credit; the OT plan is longer than it needed to be.

**WHAT SENIOR PMS DO DIFFERENTLY**

They schedule the technical reviews and their preparation, not just the test events. They treat the TRR as the real entry criterion. They involve the operational testers early and design integrated testing in the TEMP. They test to the KPP thresholds in DT so OT is not a surprise. And on the contractor side, they find their own deficiencies before the government does, because the government's test is the one that goes to Congress.

**BUILD ON THIS**

- **The Acquisition Lifecycle** The phases and milestones testing supports
- **Common Mistakes New PMs Make** Writing testable requirements and KPPs
- **Modifications, REAs & Claims** When a deficiency fix is a change

## Key Terms

### Developmental Test and Evaluation (DT&E)

Testing conducted during development to verify that the system meets its technical specifications and is ready for operational testing. Largely run by the program and the contractor, with government oversight.

### Operational Test and Evaluation (OT&E)

Testing conducted under realistic operational conditions by an independent operational test agency, using typical operators, to evaluate operational effectiveness, suitability, and survivability.

### Test and Evaluation Master Plan (TEMP)

The program's integrated plan for all testing: objectives, events, resources, schedule, and the evaluation framework linking tests to requirements. Approved at milestones.

### Director, Operational Test and Evaluation (DOT&E)

The independent OSD official who oversees operational testing on major programs, approves their OT plans, and reports to Congress on whether systems are effective and suitable.

### Technical Reviews

The gated engineering reviews: System Requirements Review (SRR), Preliminary Design Review (PDR), Critical Design Review (CDR), Test Readiness Review (TRR), and others, each with entry and exit criteria.

### Key Performance Parameters (KPPs)

The requirements a system must meet to be acceptable, from the capability document. Testing verifies them; failure to meet a KPP threshold is a program-level event.

### Operational Effectiveness and Suitability

Effectiveness: does it accomplish the mission when used by typical operators in the intended environment.

Suitability: can it be sustained in the field (reliability, maintainability, supportability, training).

### Test Readiness Review (TRR)

The review confirming that the system, the test articles, the procedures, the instrumentation, and the personnel are ready for a test event. Failing it delays the event; skipping it wastes the event.

### Integrated Testing

Designing developmental and operational testing to share events and data where possible, so that DT results inform OT and total test time and cost are reduced.

### Deficiency Report

The formal record of a test finding that the system did not perform as required, categorized by severity, with a disposition (fix, defer, accept). The output of testing that drives the contract.

## Knowledge Check

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What question does operational test and evaluation answer?

- A. Does the system meet its technical specifications?
- ✓ B. Is the system effective, suitable, and survivable in realistic use by typical operators?
- C. Is the contractor's accounting system adequate?
- D. Is the design complete?

### WHY

DT verifies specifications; OT evaluates the system in operational conditions, independently.

Who runs operational testing on a major program?

- A. The contractor
- B. The program office
- ✓ C. An independent operational test agency, with DOT&E oversight
- D. DCAA

### WHY

Independence is by law. The contractor may not operate the system during OT.

What does the Test Readiness Review confirm?

- A. That the requirements are complete
- ✓ B. That the article, procedures, instrumentation, safety, and personnel are ready for a specific test event
- C. That the design is complete
- D. That the contract is funded

### WHY

The TRR is the real entry criterion for an event. Skipping it produces failed or invalid tests.

**What is the TEMP?**

A. A temperature standard

✓ **B. The integrated plan for all testing: evaluation framework, events, resources, schedule, and roles, approved at milestones**

C. The contractor's test procedure

D. A DOT&E report

**WHY**

The TEMP links tests to requirements and is where test drives the budget and schedule.

**Which review confirms the detailed design is complete and ready to build?**

A. SRR

B. PDR

✓ **C. CDR**

D. TRR

**WHY**

The critical design review gates fabrication; failing it means test articles are built to an incomplete design.

**What is the most expensive kind of test event?**

A. A long one

✓ **B. One run before the system is ready, which consumes range time and articles and produces a deficiency everyone expected**

C. An integrated one

D. A software one

**WHY**

Programs that treat the TRR as a formality test twice.

**A deficiency report is issued after a test. What contract question follows?**

A. Whether to terminate

✓ **B. Whether the fix is within the contractor's existing obligation (specification not met) or a change (requirement ambiguous or changed)**

C. Whether to pay award fee

D. Whether to recompetete

**WHY**

Deficiency dispositions are contract actions, settled by clear specifications and test criteria.

**What is the purpose of integrated testing?**

A. To eliminate OT

✓ **B. To share events, articles, and data between DT and OT so the total test program is shorter, within the limits of OT credibility**

C. To let the contractor run OT

D. To skip the TEMP

**WHY**

Early involvement of the operational testers earns credit for DT data.

**Why does a contractor test earlier and harder than the government will?**

A. To increase cost

✓ **B. To find and fix its own deficiencies before government testing, because the government's test result goes to Congress**

C. Because the FAR requires it

D. To delay OT

**WHY**

A contractor that uses government testing to find its problems fails it.

**Match the review to its question:****CORRECT MATCHES**

|     |                                                                  |
|-----|------------------------------------------------------------------|
| SRR | Are requirements complete, understood, and allocated?            |
| PDR | Will the design approach meet requirements with acceptable risk? |
| CDR | Is the detailed design complete and ready to build?              |
| TRR | Are the article, procedures, and people ready for this event?    |

**WHY**

SRR: requirements understood; PDR: approach will meet them; CDR: design complete; TRR: ready for this event.

**Order the reviews and tests in a development program:****CORRECT ORDER**

1. System Requirements Review
2. Preliminary Design Review
3. Critical Design Review
4. Test Readiness Review
5. Developmental test and evaluation
6. Initial operational test and evaluation

**WHY**

SRR, PDR, CDR, TRR, developmental testing, then operational testing before the full-rate production decision.

**A KPP threshold fails in operational testing. What can happen?**

A. Nothing; KPPs are advisory

**✓ B. The full-rate production decision can stop, a requirements review or redesign can be required, and the failure appears in DOT&E's report to Congress**

C. The contractor is automatically terminated

D. The test is repeated by the contractor

**WHY**

A failed KPP is a program-level event. Testing to thresholds early in DT prevents the surprise.

## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 3

# The Software Acquisition Pathway and Agile Under a Government Contract

26 min

*Why DoD created a separate pathway for software, what it changes (no milestones, a minimum viable product, continuous delivery), how Agile development actually works inside a FAR contract, the authorization to operate that gates every release, the metrics that replace earned value, and where programs get it wrong.*

## Why Software Got Its Own Pathway

The major capability pathway, with its milestones, phases, and years-long development before anything is fielded, was built for hardware. Applied to software, it produced systems that were obsolete when delivered, specifications written years before anyone used the product, and testing at the end of a decade. DoD created the software acquisition pathway to do what modern software organizations do: deliver something useful quickly, learn from users, and keep delivering. It removes the milestones, replaces detailed requirements with a needs statement and a backlog, requires a minimum viable capability within a year, and expects continuous delivery afterward. This lesson explains how it works, how Agile development fits inside a government contract, and why the authorization to operate is the real gate.

## Hardware Pathway vs. Software Pathway

|                         | MAJOR CAPABILITY ACQUISITION                               | SOFTWARE ACQUISITION PATHWAY                                                             |
|-------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Structure               | Milestones A, B, C; phases with years between them         | A planning phase, then execution; no milestones                                          |
| Requirements            | Detailed capability documents validated before development | A capability needs statement; detail emerges in the backlog through user feedback        |
| First delivery to users | After development and test, often years                    | A minimum viable capability within one year of execution start                           |
| Delivery cadence        | Blocks or increments, infrequent                           | Continuous; releases measured in weeks                                                   |
| Testing                 | Discrete DT and OT campaigns                               | Continuous automated testing in the pipeline; operational evaluation of fielded releases |
| Oversight               | Milestone decisions; APB breaches                          | Annual value assessments; metrics on delivery and user outcomes                          |
| Cost management         | EVM against a baseline                                     | Cost per release, backlog burn, and value metrics; EVM generally not applied             |

## The Two Phases

The pathway begins with a planning phase: understand the users and the mission, define the capability needs, establish the product team and the development environment, choose the acquisition strategy and the contract vehicle, and plan the pipeline and the ATO approach. It ends with a decision to enter execution. The execution phase is continuous: the team builds and delivers software in sprints, users evaluate each release, the product owner reprioritizes the backlog, and the program reports on metrics and value annually. The MVCR is due within a year of entering execution. There is no Milestone C; the program keeps delivering until the capability is no longer needed. The two phases can be entered for a new program or for an existing system moving from another pathway.

## Agile Inside a FAR Contract

Agile development assumes that what gets built changes as users learn. A FAR contract assumes a defined scope, a price, and a period. Reconciling them is the practical problem, and it has working answers. The contract buys a development capacity (a team of a defined size and skill mix) for a period, usually under a cost-type, labor-hour, or fixed-price-per-sprint arrangement, with the statement of work describing the process (sprints, reviews, the definition of done, the pipeline, the quality standards) rather than the features. The government product owner directs the backlog within the contract's scope, which is described as the capability area rather than a feature list. Each sprint ends with working software the government accepts or rejects against the definition of done. What the contract does not do is fix the features in advance, because

that defeats the purpose. What it must do is bound the capability area, so that backlog direction is not a series of constructive changes.

#### IN PLAIN TERMS

Think of remodeling a kitchen by hiring a crew by the week instead of handing over a finished blueprint. Each Monday you decide what they tackle next, and you check their work every Friday. What you cannot do is send that crew to re-roof the garage, because you hired them for the kitchen. An Agile contract works the same way: the government buys a team's capacity and process, the product owner sets weekly priorities, and the capability area in the contract marks the edge of the kitchen.

#### REAL TALK

The most common failure in government Agile is a team that says "Agile" and means "we would like to change our minds constantly and also pay a fixed price." Real Agile has more discipline than waterfall, not less: one product owner who actually decides, a fixed sprint length, a definition of done, working software at the end of every sprint, and metrics somebody looks at. A program with a product owner who cannot make a call, a backlog nobody grooms, and sprints that quietly slip is not Agile. It is a cost-type contract with worse paperwork.

## What a Software Pathway Program Needs to Have

- **A real product owner** A government person with authority to prioritize and accept, available daily, who represents the users. Without this role the contractor decides what to build, which is the personal services problem in reverse.
- **A development environment and pipeline** Cloud or on-premises infrastructure, automated build and test, security scanning, deployment tooling. Often the first deliverable and the first ATO.
- **An ATO strategy** How releases will be authorized. A continuous ATO, where the pipeline's controls are accredited and releases inherit the authorization, is the goal; without it, every release waits months.
- **User access** Real users who evaluate releases and whose feedback reaches the backlog. The pathway is built on this feedback loop.
- **A contract that buys capacity and process** Team, sprints, definition of done, quality and security standards, with scope bounded by capability area.
- **Metrics from day one** Delivery and value metrics collected by the pipeline, not assembled by hand for a review.

## The ATO: The Gate That Actually Controls Delivery

Nothing runs on a government network without an authorization to operate under the Risk Management Framework: security controls implemented, assessed, and accepted by an authorizing official. Traditionally

an ATO was granted to a system version after months of assessment, which means a program delivering every two weeks would spend all its time in accreditation. The answer is a continuous ATO: the development pipeline itself is accredited (its security scanning, testing, configuration control, and deployment controls), and releases that pass through the pipeline inherit the authorization. Getting there requires the security organization's participation from the start, a pipeline built to the controls, and evidence generated automatically. Programs that treat the ATO as something to handle after development discover that their first release waits six months for it.

One sentence to teach your team: **the software pathway replaces milestones with a planning phase and continuous execution, requires a useful release within a year, buys development capacity and process under the contract with the government owning the backlog, and lives or dies on whether the pipeline can get a continuous authorization to operate.**

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#### INTERMEDIATE

### Metrics That Replace Earned Value

Earned value assumes a baseline of defined work; Agile does not have one, so the pathway uses different measures. Delivery metrics from the pipeline: deployment frequency (how often working software reaches users), lead time for changes (from commit to production), change failure rate (releases that cause problems), and time to restore service. Backlog metrics: velocity (work completed per sprint, useful for trend, not for comparing teams), burn-down against the current release plan, and backlog health (groomed, prioritized, sized). Value metrics: user adoption, user satisfaction, mission outcomes the software was meant to improve. And cost: cost per sprint, cost per release, cost per user. The annual value assessment asks whether the money is producing value, and these are the evidence. A program reporting only cost and schedule against a plan has imported the hardware model into the pathway.

## Contract Structures for Agile Development

| STRUCTURE                             | HOW IT WORKS                                                                                      | FITS WHEN                                                    | RISK                                                                                              |
|---------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Cost-plus (CPFF or CPIF) for a team   | Government pays cost plus fee for a defined team capacity; scope is the capability area           | Uncertain requirements; government wants maximum flexibility | Cost discipline depends on the product owner and metrics; fee incentives must be tied to delivery |
| Labor-hour per sprint                 | Fixed rates, hours per sprint; ceiling per period                                                 | Team size stable; government controls priorities             | T&M D&F required; surveillance of hours; ceiling management                                       |
| Fixed price per sprint or per release | A fixed price for each sprint of a defined team, or for each release meeting a definition of done | Mature team and pipeline; predictable cadence                | Definition of done must be precise; scope creep within a sprint becomes a dispute                 |
| Modular contracts and OTAs            | Short, competed increments; other transactions for prototyping                                    | Early capability; non-traditional vendors                    | Transition between increments and vendors; data and code ownership must be secured                |

## Requirements Without Requirements Documents

The pathway's capability needs statement is short and high-level, and the detail lives in the backlog. This makes some people in the requirements community nervous, and it should make the PM careful. The discipline is that the needs statement bounds the capability area and states the mission outcomes, user roles, and constraints (security, interfaces, data), and the backlog is traceable to it. A backlog item that cannot be traced to a stated need is either a new requirement (which needs the needs statement updated) or scope creep. The product owner's authority is bounded by the needs statement, which is also what bounds the contract. Programs that let the backlog drift beyond the needs statement drift beyond the contract too.

## Testing and Independent Evaluation of Software Releases

The Test and Evaluation lesson described continuous testing. In practice: automated unit, integration, and regression tests run on every build; security scans run in the pipeline; performance and load tests run before release; and users evaluate each release in an operational setting. The independent operational test agencies evaluate fielded releases and the pipeline's test rigor rather than running a single OT campaign, and DOT&E has issued guidance for software-intensive programs. The TEMP describes the continuous approach. The government still needs evidence that the software is effective and suitable; the difference is that the evidence accumulates from fielded use rather than from one event.

**ADVANCED**

## Data, Code, and the Exit

Agile development under contract produces a codebase, a pipeline, a set of environments, and operational knowledge. If the contract ends and the government does not own or control them, the next contractor starts over and the government has bought a service, not a capability. The Data Rights lesson covers the mechanics; for software the essentials are: the government owns or has unlimited or government purpose rights in the code it paid for (with the development funded at government expense and delivered continuously to a government-controlled repository, not at contract end); the pipeline and environments are government-owned or transferable; open source and commercial components are inventoried with their licenses; and the contract requires transition support. A program that discovers at recompile that the code lives in the contractor's repository with the contractor's pipeline has locked itself in as effectively as any hardware OEM.

## Governing a Software Program: What the Annual Value Assessment Should Ask

The value assessment replaces the milestone decision, and it should be as demanding. Is the software being used, by whom, for what, and is adoption growing? Are the mission outcomes it was meant to improve improving? Is the delivery cadence healthy (frequent releases, low failure rate, fast recovery)? Is the cost per release and per user reasonable and trending the right way? Is the backlog traceable to the needs statement? Is the ATO continuous and are security findings being resolved in the pipeline? Does the government own the code and the pipeline? Is the product owner effective? A program that cannot answer these with data from its own pipeline and users is not ready for the assessment, and a decision authority that does not ask them is running the hardware model with the milestones removed.

## Where Software Pathway Programs Go Wrong

The recurring patterns.

### No real product owner **GOVERNANCE**

A part-time government lead who cannot prioritize or accept; the contractor fills the vacuum.

Fix: a full-time, empowered product owner with user access, or the program is not ready for execution.

### ATO as an afterthought **SECURITY**

Pipeline built without the security organization; first release waits months for accreditation.

Fix: the ATO strategy and the pipeline controls are planning-phase deliverables.

### Waterfall in sprints **METHOD**

Requirements fixed up front, sprints used only as a schedule, no user feedback until the end.

Fix: an MVCR to real users within the year, and a backlog that changes because of what they say.

**Metrics assembled by hand** DATA

Delivery metrics compiled for reviews rather than generated by the pipeline.

Fix: instrument the pipeline in the planning phase; if the metrics are not automatic, the pipeline is not done.

**Code in the contractor's repository** OWNERSHIP

Continuous delivery to an environment the government does not control.

Fix: government-controlled repository and pipeline from day one, with delivery every sprint.

**WHAT SENIOR PMS DO DIFFERENTLY**

They staff a real product owner before execution starts. They treat the pipeline and the continuous ATO as the first deliverables. They buy capacity and process under a contract bounded by a needs statement. They deliver to real users within the year and change the backlog because of what those users say. They generate metrics from the pipeline. And they keep the code and the pipeline in government hands from the first sprint.

**BUILD ON THIS**

- **The Acquisition Lifecycle** The six AAF pathways
- **Data Rights and IP** Owning the code you paid for
- **CMMC and DFARS 7012** Security requirements the pipeline must meet

## Key Terms

### Software Acquisition Pathway

One of the six Adaptive Acquisition Framework pathways, created for software-intensive systems. Replaces milestones with a planning phase and an execution phase, requires a minimum viable capability within a year of execution start, and expects continuous delivery afterward.

### Minimum Viable Product (MVP) / Minimum Viable Capability Release (MVCR)

The smallest useful increment delivered to users early, to learn from real use and to prove the delivery pipeline. The pathway expects an MVCR within one year of starting execution.

### Agile

An iterative development approach: short cycles (sprints), a prioritized backlog of work, frequent delivery of working software, and continuous feedback from users. The method the pathway assumes.

### DevSecOps

Development, security, and operations integrated in one automated pipeline: code is built, tested, security-scanned, and deployed continuously rather than in phases.

### Authorization to Operate (ATO)

The formal accreditation that a system is secure enough to operate on government networks, under the Risk Management Framework. Every release needs it; a continuous ATO lets a pipeline release without re-accrediting each time.

### Product Owner

The government role that owns the backlog: decides what the development team builds next, in what priority, and accepts what is delivered. The pathway's answer to the requirements problem.

### Capability Needs Statement

The pathway's requirements document: high-level needs rather than detailed specifications, with the detail emerging through the backlog and user feedback.

### Sprint

A fixed-length development cycle, typically two to four weeks, ending with working software the product owner can accept or reject.

### Value Assessment

The pathway's annual review of whether the software delivered is providing the expected value to users, replacing the milestone decisions of the hardware pathway.

### Software Metrics

Measures of delivery health: deployment frequency, lead time for changes, change failure rate, time to restore, plus backlog burn-down, user satisfaction, and cost per release. The pathway's replacement for earned value.

## Knowledge Check

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What replaces milestones in the software acquisition pathway?

- A. Nothing; milestones remain
- ✓ B. A planning phase followed by continuous execution, with annual value assessments
- C. A single Milestone C
- D. Quarterly DCAA audits

### WHY

The pathway has no Milestones A, B, or C. Execution continues until the capability is no longer needed.

When must a minimum viable capability be delivered to users?

- A. At the end of the contract
- ✓ B. Within one year of entering execution
- C. Within five years
- D. Before the planning phase

### WHY

The MVCR requirement forces early delivery and early user feedback.

What does a contract under the software pathway typically buy?

- A. A fixed list of features
- ✓ B. Development capacity and process (team, sprints, definition of done, pipeline, standards), with scope bounded by a capability area
- C. Only the pipeline
- D. Hardware

### WHY

Fixing features in advance defeats the purpose; bounding the capability area keeps backlog direction inside the contract.

**Who owns the backlog on a software pathway program?**

A. The contractor's scrum master

✓ B. The government product owner, who prioritizes and accepts

C. The contracting officer

D. DOT&E

**WHY**

Without an empowered government product owner, the contractor decides what to build.

**What is a continuous ATO?**

A. An ATO that never expires

✓ B. Accreditation of the pipeline's controls so that releases passing through it inherit the authorization without re-accrediting each one

C. An ATO granted by the contractor

D. A waiver of security requirements

**WHY**

Without it, every release waits months for accreditation and continuous delivery is impossible.

**Which metrics replace earned value on the pathway?**

A. CPI and SPI

✓ B. Delivery metrics (deployment frequency, lead time, change failure rate, time to restore), backlog metrics, value metrics, and cost per release

C. Only cost

D. Milestone dates

**WHY**

Agile has no fixed baseline for EVM. Pipeline-generated metrics and user outcomes are the evidence.

**A program calls itself Agile but has fixed requirements, sprints used only as a schedule, and no user feedback until the end. What is it?**

A. Agile

✓ **B. Waterfall in sprints**

C. A software pathway program

D. DevSecOps

#### WHY

Agile requires a changing backlog driven by user feedback and working software every sprint.

**What bounds the product owner's authority and the contract's scope?**

A. The sprint length

✓ **B. The capability needs statement, to which every backlog item should be traceable**

C. The ATO

D. The velocity

#### WHY

A backlog item not traceable to a stated need is either a new requirement or scope creep.

**At recompete, the code lives in the contractor's repository with the contractor's pipeline. What has happened?**

A. Normal practice

✓ **B. The government has locked itself in as effectively as a hardware OEM; it bought a service, not a capability**

C. The contractor owns the software

D. Nothing; code transfers automatically

#### WHY

Government-controlled repository and pipeline from the first sprint, with delivery every sprint, prevents this.

**Match the contract structure to its fit:****CORRECT MATCHES**

|                                   |                                                      |
|-----------------------------------|------------------------------------------------------|
| Cost-plus for a defined team      | Uncertain requirements; maximum flexibility          |
| Labor-hour per sprint             | Stable team; government controls priorities; T&M D&F |
| Fixed price per sprint or release | Mature team and pipeline; precise definition of done |
| Modular contracts and OTAs        | Early capability; non-traditional vendors            |

**WHY**

Cost-plus for uncertain requirements; labor-hour for stable team with government priorities; fixed price per sprint for mature teams; OTAs for early prototyping.

**Order a software pathway program's early steps:****CORRECT ORDER**

1. Define users, mission, and the capability needs statement
2. Staff the product owner and build the development environment
3. Plan the pipeline and the continuous ATO with the security organization
4. Award a contract for capacity and process bounded by the needs statement
5. Enter execution and deliver an MVCR within one year

**WHY**

Planning phase: users and needs, product team and environment, ATO strategy and pipeline, contract; then execution with an MVCR within a year.

**What should an annual value assessment ask?**

- A. Whether the milestone was met
- ✓ B. Whether the software is used, whether mission outcomes are improving, whether delivery cadence and cost per release are healthy, whether the backlog traces to needs, whether the ATO is continuous, and whether the government owns the code**
- C. Only whether the budget was spent
- D. Whether EVM indices are above 1.0

**WHY**

The value assessment replaces the milestone decision and should be as demanding, with data from the pipeline and users.

## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 4

# Contract Closeout: The Work After the Work

22 min

*A contract is not over when the last deliverable is accepted. The closeout process, why cost-type contracts take years to close, what is stuck until they do (fee, final rates, cash), quick closeout, the releases and certifications, property disposition, records retention, and how both sides get it done instead of letting it pile up.*

## The Contract Is Not Over

The last deliverable is accepted, the team moves to the next program, and everyone assumes the contract is done. It is not. Closeout is a defined process with its own steps, and until it finishes, money is stuck on both sides: the contractor cannot collect withheld fee or settle its final costs, and the government cannot deobligate unspent funds or close the file. On fixed-price contracts closeout is quick. On cost-type contracts it routinely takes years, because it waits for final indirect rates that wait for incurred cost audits. Contracting offices carry backlogs of thousands of physically complete contracts, and contractors carry balance sheets full of unsettled years. The Cash Is Not Revenue lesson called this the closeout cash trap. This lesson explains the process and how to move it.

### IN PLAIN TERMS

Think about moving out of a rental apartment. You hand back the keys, but your security deposit stays with the landlord until the place is inspected, and the final utility bill shows up weeks later. Until both are settled, neither side is done. Contract closeout works the same way: the work is finished, but withheld fee, final indirect rates and unspent government funds stay stuck until every step of closeout is completed.

## Closeout Timelines the FAR Expects (FAR 4.804)

| CONTRACT TYPE                                    | TARGET FOR CLOSEOUT AFTER PHYSICAL COMPLETION              | WHY                                                        |
|--------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| Simplified acquisitions                          | When the CO receives evidence of receipt and final payment | Nothing to settle                                          |
| Firm-fixed-price (other than simplified)         | 6 months                                                   | Final invoice, property, releases; no rate settlement      |
| Contracts requiring settlement of indirect rates | 36 months                                                  | Waits for final rates for every fiscal year of performance |
| All others                                       | 20 months                                                  | Intermediate cases                                         |

## Why Cost-Type Contracts Take Years

A cost-type contract bills indirect costs at provisional rates during performance. The final rates for each fiscal year are established only after the contractor submits its incurred cost proposal (due six months after the fiscal year ends) and DCAA audits it or the ACO negotiates it. A contract that performed across four fiscal years needs four years of final rates, and the last year's rates cannot be final until at least a year and a half after the contract ends, often much longer if the audit is queued. Until then, the contractor cannot submit its final voucher, the government cannot make final payment or release withheld fee, and the contract cannot close. Every cost-type contract the company performs is in this pipeline, which is why contractors with weak incurred cost submissions have years of open contracts and cash they cannot reach.

## The Closeout Checklist

The steps, in roughly the order they happen. Most contracting offices use a checklist form; the contents are stable.

- **Confirm physical completion** All deliverables accepted, period of performance ended, no open options. Documented by the COR.
- **Disposition of government property** Inventory reconciled; property returned, transferred, or disposed of; the property administrator concurs. Unaccounted property is a finding and a delay.
- **Classified material and security** Returned or destroyed per the DD 254; personnel debriefed; the FSO certifies.
- **Subcontract closeout** The prime closes its subcontracts, settles their costs, and obtains their releases. A prime cannot close until its subs have.
- **Final indirect rates (cost-type)** Settled for every fiscal year of performance, or quick closeout applied.
- **Final invoice or voucher** Reconciling all costs, fee, and prior payments; releasing withholds.
- **Contractor releases and certifications** Release of claims; assignment of refunds, rebates, and credits; any other required certifications.
- **Patent and data deliverables** Final invention reports and technical data delivered with correct markings.
- **Deobligation and the closeout modification** Unspent funds removed; the contract administratively closed by modification; the file marked closed.

**REAL TALK**

Closeout is nobody's job, which is exactly why it is always late. The PM moved on, the COR has a new contract, the contractor's finance team is chasing invoices that might actually get paid this month, and the contracting office is measured on awards, not on finishing things. So the withheld fee sits there, the expiring funds quietly cancel, and the stack of unclosed years grows. The contractor who assigns one person, a tracker, and a standing monthly call with the ACO closes contracts years ahead of everyone waiting to be asked. It is the least glamorous competitive advantage in the business.

## Quick Closeout: The Procedure That Exists to Prevent the Backlog

FAR 42.708 lets the CO settle a contract's indirect costs before final rates are established when the contract is physically complete, the amount of unsettled indirect cost is relatively small (below a dollar threshold or a percentage of the contract's total cost), and the CO can negotiate a settlement using the best available information (provisional rates, the most recent final rates, or the contractor's proposed rates adjusted). The settlement is final for that contract and does not affect other contracts' rates. Quick closeout is underused because it requires someone to ask for it. A contractor with a physically complete cost-type contract and modest unsettled indirect cost should propose quick closeout to the ACO; a government office with a backlog should offer it.

**EXAMPLE (ILLUSTRATIVE)**

A 60-person engineering firm finished a \$6,000,000 cost-type contract that ran across four of its fiscal years. Final indirect rates are settled for the first three years. The last year is still waiting in the audit queue, so the contract cannot close and the firm's withheld fee sits unpaid. The firm calculates the indirect cost still unsettled on this contract at about \$80,000. That is  $\$80,000 \div \$6,000,000$ , or about 1.3 percent of the contract's total cost. The firm writes to the administrative contracting officer (ACO) proposing quick closeout. The ACO confirms the amount is within the quick closeout limits, negotiates using the most recent final rates, and settles. The final voucher goes in, the fee is released, and the contract closes years early.

One sentence to teach your team: **physical completion starts closeout rather than ending the contract, cost-type contracts wait years for final rates unless quick closeout is used, and money is stuck on both sides until the releases are signed, the property is accounted for, and the closeout modification is issued.**

## INTERMEDIATE

## What Is Stuck Until Closeout

On the contractor side: withheld fee (on CPFF contracts, up to 15% of fee, capped, held until closeout requirements are met), the difference between provisional and final rates (owed to the contractor if final rates are higher), any amounts held under business system withholds, and the final voucher's reconciliation. On the balance sheet, unbilled receivables and contract assets that cannot convert to cash, which the Cash lesson explained acquirers notice. On the government side: obligated funds that cannot be deobligated and reused, which matter most when they are about to expire or cancel, and open contract files that consume administrative attention and appear in audit findings about closeout backlogs. Both sides have money in the same stuck place, which should make closeout a shared priority. It rarely is.

## The Release of Claims: Read It Before Signing

The release of claims is the contractor's statement that it has no further claims under the contract, except those listed. Signing it without listing an open request for equitable adjustment, an unresolved change, or a pending dispute waives them. The correct practice is to list every open item specifically, with amounts where known, as exceptions to the release. The government cannot close a contract with unresolved exceptions, which is the point: the exceptions force resolution. A contractor that signs a clean release to speed closeout has traded its claims for the withheld fee, and the trade is usually bad.

## Closeout Roles

| ROLE                             | RESPONSIBILITY                                                                                                             |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Contracting officer (PCO or ACO) | Owns closeout; issues the closeout modification; negotiates quick closeout; deobligates funds                              |
| Contractor contracts and finance | Final voucher; releases and certifications; subcontract closeout; incurred cost submissions that make final rates possible |
| COR                              | Confirms physical completion and acceptance; provides the performance file                                                 |
| Property administrator (DCMA)    | Concurs on property disposition                                                                                            |
| DCAA                             | Audits incurred cost submissions; establishes final rates (or the ACO negotiates)                                          |
| FSO and security                 | Classified material and personnel actions per the DD 254                                                                   |
| Resource manager                 | Tracks obligations for deobligation and expiring funds                                                                     |

## Subcontract Closeout: The Step Primes Forget

A prime cannot close a contract until its subcontracts under it are closed, which means settling each sub's final costs, obtaining each sub's release of claims, and disposing of any property the sub held. On a program with a dozen subs, some of them small firms that no longer exist, this is the step that stalls. Primes that close subcontracts as each one finishes, rather than at the end of the prime contract, avoid the problem. Primes that include closeout obligations, releases, and a final invoice deadline in their subcontract terms have the leverage to finish.

### ADVANCED

## Running a Closeout Program at a Contractor

Contractors with many cost-type contracts run closeout as a program. An inventory of every physically complete contract with its fiscal years, the status of final rates for each year, the amounts withheld and unbilled, the open subcontracts, and the open exceptions. A priority order by cash value. Incurred cost submissions filed on time every year, adequate on the first submission, because every late or inadequate submission delays final rates on every contract that performed that year. Quick closeout proposed on every eligible contract. A monthly review with the ACO. And a named owner. The output is cash: withheld fee collected, rate differences settled, the balance sheet cleaned up before a sale or a bank review. The Business module's valuation lesson noted that open audit years become escrows in a sale; a closeout program is how they are avoided.

## Closeout and the Audit Trail

Closeout is also the point at which the contract file becomes the record. Records must be retained for the statutory period after final payment (generally three years, longer for certain categories), and the file is what an inspector general, a DCAA post-award audit, or a False Claims Act investigation reads. A well-closed contract has a complete file: the acceptance documentation, the property records, the final rates and voucher, the releases with their exceptions, the deobligation, and the closeout modification. A badly closed one has a modification and a shrug. Senior PMs on both sides treat the closeout file as the last deliverable, because it is the one that will be read years later by people who were not there.

## Closeout Accelerators

What moves a stuck contract.

### Propose quick closeout RATES

For any physically complete cost-type contract with modest unsettled indirect cost.

The CO can settle on the best available information. Ask.

**File incurred cost submissions on time and adequate** RATES

Six months after fiscal year end, complete, using the standard model.

Late or inadequate submissions hold every contract that performed that year.

**Close subcontracts as they finish** SUBS

Do not wait for the prime contract's end.

Subcontract terms should require a final invoice and release within a set period.

**Reconcile property continuously** PROPERTY

Annual inventories that match the property system; disposition requests filed as items become excess.

Property discrepancies at closeout are findings and months of delay.

**List the exceptions and settle them** CLAIMS

Open REAs and changes listed on the release, then resolved.

The exception forces the resolution; a clean release forfeits the claim.

**WHAT SENIOR PMS DO DIFFERENTLY**

They treat physical completion as the start of a process with an owner. They file incurred cost submissions on time because every contract depends on them. They propose quick closeout on every eligible contract. They close subcontracts as they finish and reconcile property as they go. They list every open claim on the release. And they know the cash value of their closeout backlog, because it is money the company earned and cannot touch.

**BUILD ON THIS**

- **Cash Is Not Revenue** The closeout cash trap and provisional rates
- **DCAA & DCMA Oversight** Incurred cost audits and final rates
- **Modifications, REAs & Claims** The claims that must be excepted from the release

## Key Terms

### Physical Completion

The point at which the contractor has delivered everything and the government has accepted it, or the period of performance has ended. The start of closeout, not the end of the contract.

### Administrative Closeout

The process after physical completion: final invoicing, rate settlement, releases, property disposition, records, and the closeout modification. Ends with the contract file being closed.

### Final Indirect Rates

The audited or negotiated indirect rates for each fiscal year of performance on a cost-type contract, established through the incurred cost submission and DCAA audit. Required before the final voucher.

### Final Voucher

The last invoice on a cost-type contract, submitted after final rates are settled, reconciling all costs and fee and releasing any withheld amounts.

### Quick Closeout

A FAR procedure (42.708) allowing the CO to settle indirect costs on a contract before final rates are established, when the unsettled amount is small, so the contract can close without waiting years.

### Release of Claims

The contractor's certification that it has no further claims against the government under the contract, except any specifically excepted. Required for closeout; it ends the contractor's ability to raise new claims.

### Assignment of Refunds, Rebates, and Credits

The contractor's assignment to the government of any refunds or credits later received that relate to costs the government reimbursed.

### Government Property Disposition

Accounting for and returning, transferring, or disposing of all government-furnished and contractor-acquired property, with the property administrator's concurrence.

### Deobligation

Removing unspent obligated funds from the contract at closeout, returning them to the government for other use before they expire or cancel.

### Records Retention

The requirement to keep contract records for a set period after final payment (generally three years, longer for some records), available for audit.

## Knowledge Check

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What does physical completion mean?

- A. The contract is closed
- ✓ B. All deliverables are delivered and accepted or the period of performance has ended; closeout begins
- C. Final payment has been made
- D. The COR has been released

### WHY

Physical completion starts administrative closeout; it does not end the contract.

What is the FAR's target for closing a contract that requires settlement of indirect rates?

- A. 6 months
- B. 20 months
- ✓ C. 36 months
- D. 10 years

### WHY

FAR 4.804 sets 36 months for contracts needing rate settlement, 6 months for fixed-price, 20 months for others.

Why do cost-type contracts take years to close?

- A. Because deliverables are late
- ✓ B. Because final indirect rates for every fiscal year of performance must be established through incurred cost submissions and audits before the final voucher
- C. Because of protests
- D. Because CORs delay them

### WHY

The last year's rates cannot be final until well after the contract ends, and audits queue.

**What is quick closeout?**

A. A fixed-price closeout

✓ **B. A FAR procedure allowing the CO to settle indirect costs before final rates are established, when the unsettled amount is small**

C. A closeout without releases

D. An automatic closeout after 36 months

**WHY**

FAR 42.708. Underused because someone has to ask for it.

**A contractor signs a clean release of claims while an REA is pending. What happens?**

A. The REA continues

✓ **B. The REA is waived; claims not listed as exceptions are released**

C. The government pays the REA at closeout

D. Nothing until final payment

**WHY**

List every open item as an exception. The exception forces resolution.

**What is stuck on the contractor's side until a cost-type contract closes?**

A. Nothing

✓ **B. Withheld fee, the provisional-to-final rate difference, business system withholds, and the final voucher reconciliation**

C. Only the final invoice

D. Award fee only

**WHY**

Cash the company earned and cannot collect, which also appears as unsettled contract assets to acquirers.

**Why can a prime not close a contract until its subcontracts are closed?**

A. It can

✓ **B. Because sub costs must be settled and sub releases obtained before the prime's final voucher and release are complete**

C. Because DCAA audits subs first

D. Because of property rules

**WHY**

Closing subcontracts as they finish, with closeout terms in the subcontract, prevents the stall.

**Which single practice most affects how quickly a contractor's cost-type contracts close?**

A. Fast invoicing

✓ **B. Filing incurred cost submissions on time and adequate every year**

C. Hiring more CORs

D. Using fixed-price subs

**WHY**

Every late or inadequate submission delays final rates on every contract that performed that year.

**What happens to unspent obligated funds at closeout?**

A. They are paid to the contractor

✓ **B. They are deobligated and returned for other use before they expire or cancel**

C. They stay on the contract forever

D. They become fee

**WHY**

Deobligation is the government's financial interest in closing contracts promptly.

**Match the closeout role to its responsibility:****CORRECT MATCHES**

|                        |                                                                          |
|------------------------|--------------------------------------------------------------------------|
| Contracting officer    | Issues the closeout modification; negotiates quick closeout; deobligates |
| COR                    | Confirms physical completion and acceptance                              |
| DCAA                   | Audits incurred cost and establishes final rates                         |
| Property administrator | Concurs on government property disposition                               |

**WHY**

The CO owns closeout; the COR confirms completion; DCAA sets final rates; the property administrator concurs on disposition.

**Order the closeout steps for a cost-type contract:****CORRECT ORDER**

1. Confirm physical completion
2. Dispose of government property and close security
3. Close subcontracts and obtain their releases
4. Settle final indirect rates (or quick closeout)
5. Submit the final voucher
6. Sign releases and certifications with exceptions listed
7. Deobligate funds and issue the closeout modification

**WHY**

Physical completion, property and security, subcontract closeout, final rates, final voucher, releases, deobligation and closeout modification.

**Why does a closeout backlog matter in a company sale?**

A. It does not

✓ **B. Open audit years and unsettled contracts become escrows and holdbacks, and stuck cash reduces the working capital delivered**

C. It increases the multiple

D. Buyers prefer open contracts

**WHY**

The valuation lesson: open items become deal terms that move money away from the seller.

## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 5

# Terminations: Convenience, Default, and What Each Costs

25 min

*The two ways a contract ends early. Termination for convenience, the government's right to walk away and what the contractor can recover; termination for default, the cure notice and show cause that precede it, excess reprocurement costs, and the conversion that turns a default into a convenience. How each side prepares, responds, and settles.*

## Two Ways to End Early, and They Are Not Alike

Government contracts contain two termination clauses that sound similar and could not be more different in effect. Termination for convenience lets the government stop for its own reasons, with the contractor made whole for the work done and the cost of stopping. Termination for default punishes the contractor for failing to perform: it gets paid only for what was accepted and may owe the government the extra cost of finding someone else. Both are rare relative to the number of contracts, and both are the events a PM on either side most needs to understand before they happen, because the response in the first days determines the outcome. This lesson covers both, from the notice to the settlement.

### IN PLAIN TERMS

Picture hiring a painter for five rooms. If you change your mind after two rooms, you pay for the two rooms done and the paint already bought, but not the profit on rooms never painted. If the painter simply stops showing up, you pay only for rooms you accept, hire someone pricier to finish, and send the first painter the difference. The first is termination for convenience. The second is termination for default.

## Convenience vs. Default

|                         | TERMINATION FOR CONVENIENCE                                                                                                                      | TERMINATION FOR DEFAULT                                                                                   |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Why                     | In the government's interest: requirement eliminated, funding lost, technology overtaken, program restructured                                   | Contractor failure: no delivery, no progress, noncompliance                                               |
| Process                 | Written notice; stop work; settlement proposal; negotiation                                                                                      | Cure notice or show cause; contractor response; CO decision; notice of default                            |
| Contractor recovers     | Costs incurred on the work, settlement expenses, profit on work performed (no anticipated profit on unperformed work), subcontractor settlements | Payment for accepted deliverables only; unliquidated progress payments may be recovered by the government |
| Contractor owes         | Nothing                                                                                                                                          | Excess reprocurement costs; possibly liquidated damages                                                   |
| Past performance effect | Neutral; not a performance failure                                                                                                               | Severe; reported in CPARS and FAPIIS; affects responsibility on future awards                             |
| Contractor's remedy     | Dispute the settlement amount                                                                                                                    | Challenge the default (excusable delay, waiver, improper process); if improper, converts to convenience   |

### Termination for Convenience: The Government's Right to Walk Away

The convenience clause lets the government end a contract whenever the CO determines it is in the government's interest. Courts have limited it only at the edges (it cannot be used in bad faith or as a pretext to get a better price elsewhere). The reasons are usually mundane: the requirement went away, the funding did not arrive, a program was cancelled, a better solution appeared. On receipt of the notice, the contractor must stop work as directed, terminate subcontracts, protect government property, and begin the settlement process. The government owes the contractor the costs it incurred on the terminated work, the costs of settling (including subcontractor settlements and the cost of preparing the proposal), and a reasonable profit on the work performed. It does not owe anticipated profit on the work that will never be done. On a fixed-price contract, if the contractor would have lost money, the settlement is reduced accordingly (the loss adjustment). Total recovery cannot exceed the contract price.

**EXAMPLE (ILLUSTRATIVE)**

A 30-person firm holds a \$2,000,000 firm-fixed-price contract. The program is cancelled when the work is about 40 percent done, and the government terminates for convenience. The firm stops work that day and opens new charge numbers for termination effort. Its settlement proposal includes \$750,000 of costs incurred on the terminated work, not previously paid; \$60,000 of profit on work performed (8 percent of \$750,000, a hypothetical negotiated rate); \$40,000 of settlement expenses for accounting and proposal preparation; and \$50,000 to settle a subcontractor. Total:  $\$750,000 + \$60,000 + \$40,000 + \$50,000 = \$900,000$ . That is under the \$2,000,000 contract price, so the cap does not bite. Profit on the 60 percent of work never performed is not recoverable.

## The Contractor's First Week After a Convenience Termination

- **Stop work as directed** Immediately, on the terminated scope, and document the stop: what was in progress, what was finished, what was ordered.
- **Notify subcontractors** In writing, with the same effective date, and start their settlements.
- **Segregate costs** Open new charge numbers for termination effort so settlement expenses are captured separately from the work.
- **Inventory and protect property** Government property, work in process, materials; nothing moves without the TCO's direction.
- **Preserve the records** Everything supporting the costs incurred; the settlement will be audited.
- **Plan the settlement proposal** The form, the basis, the one-year deadline, and who at the company owns it.

**THE SETTLEMENT IS THE LAST DELIVERABLE**

A convenience termination is a business event, not a disaster, if the contractor handles it as a project. The companies that lose money on them are the ones that keep working for a week hoping for a reversal, mix termination costs into the regular charge numbers, forget a subcontractor, and submit a proposal eleven months later that the auditor cannot follow. The settlement is the last deliverable on the contract. Run it like one.

## Termination for Default: The Process Before the Decision

The government cannot default a contractor on a whim; the default clause and case law require a process. For failure to deliver on time, the government may terminate without a cure period, but it must act promptly or risk waiving the delivery date. For failure to make progress that endangers performance, or failure to comply with other terms, the government must issue a cure notice giving the contractor (usually) ten days to cure or explain. When there is not enough time or the delivery date has already passed, the government issues a show cause notice, requiring the contractor to explain why it should not be defaulted.

The contractor's response to either notice is the most important document it will write on the contract: it should state the facts, identify any excusable delay or government-caused problems, present the recovery plan with dates, and request the government's cooperation on anything it controls. A default issued without following the process, or in the face of excusable delay, is improper and will be converted.

## What Default Costs

A defaulted contractor is paid for accepted work only. Unaccepted work in process is the contractor's loss. The government may recover unliquidated progress payments. The government reprocures the work and charges the contractor the excess cost: if the defaulted price was \$10M and the replacement costs \$13M, the contractor owes \$3M, subject to the government having acted reasonably in reprocurring. The default is reported in the government's past performance systems and must be disclosed in future proposals, where it becomes a responsibility question. For a small company, a single default can be fatal. For any company, the cost of contesting an improper default is small next to the cost of accepting one.

One sentence to teach your team: **convenience termination pays the contractor for work done and the cost of stopping, default termination pays only for accepted work and can charge the contractor the cost of reprocurement, and the government must follow the cure and show cause process or the default converts to convenience.**

### INTERMEDIATE

## Excusable Delay and the Defenses to Default

The default clause excuses delays from causes beyond the contractor's control and without its fault or negligence, listing examples: acts of God or the public enemy, government acts in its sovereign or contractual capacity, fires, floods, epidemics, quarantines, strikes, freight embargoes, unusually severe weather. Subcontractor delays are excusable only if beyond the control of both the prime and the sub and the supplies were not obtainable elsewhere. Beyond excusable delay, defaults are challenged on waiver (the government let the delivery date pass without action and then tried to default), on government-caused delay (late GFP, defective specifications, constructive changes), on the contractor's reasonable prospect of performing, and on process (no cure notice when one was required). A contractor facing default assembles these defenses in its response to the notice, with documentation, because the response is the record the board of contract appeals will read.

## The Settlement Proposal: Building It

After a convenience termination, the contractor submits a termination settlement proposal within one year, on the standard forms (inventory basis for most; total cost basis when the inventory approach is not practicable). It includes: the costs incurred on the terminated work not previously paid, settlement expenses

(accounting, legal, clerical, and the cost of preparing the proposal), the cost of settling subcontracts, profit on work performed (a percentage negotiated with the TCO, not the contract's fee rate automatically), and disposition of inventory (the government may take it, direct its sale, or let the contractor keep it with a credit). DCAA audits proposals above a threshold. The TCO negotiates and the settlement is a modification. A proposal built from segregated charge numbers and a clean inventory settles in months; one reconstructed from mixed records settles in years, if at all.

## What Is and Is Not Recoverable After a Convenience Termination

| ITEM                                   | RECOVERABLE? | NOTE                                                                                                                       |
|----------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------|
| Costs incurred on terminated work      | Yes          | Allowable costs under FAR Part 31, not previously paid                                                                     |
| Profit on work performed               | Yes          | Reasonable, negotiated; reduced or eliminated if the contract would have been a loss                                       |
| Anticipated profit on unperformed work | No           | The government does not pay for what it did not receive                                                                    |
| Settlement expenses                    | Yes          | Proposal preparation, accounting, legal, subcontract settlement costs                                                      |
| Subcontractor settlements              | Yes          | The prime settles with subs and includes the amounts                                                                       |
| Unabsorbed overhead                    | Sometimes    | Recoverable under specific conditions (a recognized formula) when the termination or a suspension left overhead unabsorbed |
| Costs of continuing after the notice   | No           | Work after the stop date is the contractor's cost unless directed                                                          |
| Total above the contract price         | No           | Recovery is capped at the contract price                                                                                   |

## The Government Side: When and How to Terminate

For a government PM, a convenience termination is usually a budget or requirement decision made above the program, and the PM's job is to time it (partial terminations at logical break points cost less than mid-task stops), to plan the transition of any continuing need, and to support the TCO with acceptance records and property information. A default is the PM's and CO's decision together, and it should be the last step after the QASP's remedies, the cure process, and an honest assessment of whether the government contributed to the failure. Defaulting a contractor that has a good excusable delay defense produces a conversion, a settlement, and a reprocurement the government pays for entirely. The CO makes the call; the PM provides the performance record that makes it defensible or not.

**ADVANCED**

## Contesting a Default

A contractor that believes a default is improper appeals the CO's decision to the appropriate board of contract appeals or the Court of Federal Claims, under the Contract Disputes Act. The government bears the initial burden of showing the default was justified; the contractor then shows excusable delay or another defense. If the contractor prevails, the default is converted to a termination for convenience and the contractor recovers under those rules, including settlement expenses. The Modifications, REAs and Claims lesson covers the disputes process. The practical point is that the record made during performance and in the response to the cure or show cause notice is what the board reads; a contractor that documented government-caused delays as they happened, gave the notices its contract required, and responded to the cure notice with facts and a plan has a case. One that argued for the first time after the default does not.

## Terminations Under Vehicles, Cost-Type Contracts, and Commercial Items

The mechanics vary by contract form. Under multiple-award vehicles, the government usually terminates or simply does not exercise task orders rather than the vehicle. On cost-type contracts, convenience termination settlement is simpler because costs are already reimbursed as incurred; the settlement covers fee on work performed and settlement expenses. For commercial item contracts, FAR Part 12 has its own termination provisions: convenience termination pays a percentage of the price reflecting the work performed plus reasonable charges resulting from the termination, and default is termination for cause with a simpler process. Contractors should know which termination clause is in each contract, because the recovery rules and the process differ.

## Preparing for a Termination You Do Not Expect

The practices that make either kind of termination survivable, adopted before any notice arrives.

### Charge number discipline **COST**

Costs recorded by task, so terminated scope can be separated from continuing scope.

A partial termination on a contract with one charge number is a reconstruction project.

### Government-caused delay file **DEFENSE**

Every late GFP, late decision, and directed change documented and noticed when it happened.

The excusable delay defense is built during performance, not after the notice.

### Subcontract termination terms **SUBS**

Flow-down of the termination clauses, with the sub's settlement rights matching the prime's.

A prime cannot recover from the government what it owes a sub under terms more generous than the government's.

**Property records that are current** **PROPERTY**

Government property and work in process inventoried and reconciled.

Termination inventory schedules are built from these.

**A response template for cure notices** **PROCESS**

Facts, excusable causes, government dependencies, recovery plan with dates, request for cooperation.

Ten days is short. The structure should exist before the notice does.

**WHAT SENIOR PMS DO DIFFERENTLY**

Contractor: they stop on the date, segregate the costs, settle the subs, and submit a clean proposal early; and facing default, they respond to the cure notice with facts, defenses, and a plan, because that response is the case. Government: they time convenience terminations at break points and support the TCO; and they default only after the process, the QASP remedies, and an honest look at their own contribution, because an improper default costs the government the reprocurement and the settlement both.

**BUILD ON THIS**

- **Contract Administration & COR** Cure notices, show cause, and the escalation ladder
- **Modifications, REAs & Claims** The disputes process for contesting a default
- **Contract Closeout** Settling and closing after a termination

## Key Terms

### Termination for Convenience (T4C)

The government's contractual right to end a contract, in whole or in part, when it is in the government's interest, without the contractor being at fault. The contractor is paid for work done plus reasonable costs of stopping, plus profit on work performed.

### Termination for Default (T4D)

Ending a contract because the contractor failed to perform: failed to deliver, failed to make progress endangering performance, or failed to comply with other provisions. The contractor is paid only for accepted work and may owe the government's excess costs of reprocurement.

### Cure Notice

The written notice, required before default for failure to make progress or comply, giving the contractor a period (usually 10 days) to cure the failure or show why it should not be defaulted.

### Show Cause Notice

A notice used when there is not enough time for a cure period or after a delivery date has passed, requiring the contractor to show why the contract should not be terminated for default.

### Termination Settlement Proposal

The contractor's claim for costs after a convenience termination: costs incurred, settlement expenses, profit on work performed, and subcontractor settlements, on a standard form, due within one year.

### Excess Reprocurement Costs

After a default, the difference between what the government pays another contractor to complete the work and the defaulted contract price, chargeable to the defaulted contractor.

### Excusable Delay

Delay from causes beyond the contractor's control and without its fault or negligence (acts of God, government acts, epidemics, strikes, unusually severe weather). A defense to default.

### Conversion

When a termination for default is found improper (the failure was excusable, or the government did not follow the process), it is converted to a termination for convenience, and the contractor recovers under the convenience rules.

### Termination Contracting Officer (TCO)

The CO assigned to settle a termination, often from DCMA, who negotiates the settlement and audits the proposal.

### Partial Termination

Terminating part of a contract's scope while the rest continues. Common in convenience terminations when a requirement is reduced.

## Knowledge Check

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**Under a termination for convenience, what does the contractor recover?**

- A. The full contract price
- ✓ **B. Costs incurred on the work, settlement expenses, and reasonable profit on work performed, capped at the contract price**

- C. Anticipated profit on all remaining work
- D. Nothing

### WHY

No anticipated profit on unperformed work; recovery is capped at the contract price and reduced if the contract would have been a loss.

**Before terminating for failure to make progress, the government must generally issue:**

- A. A show cause notice only
- ✓ **B. A cure notice giving the contractor a period, usually 10 days, to cure or explain**

- C. A stop-work order
- D. A J&A

### WHY

The cure notice is required for failure to make progress or comply. Skipping it makes the default improper.

**After a default, the government reprocures at \$13M against a defaulted price of \$10M. What may the government charge the contractor?**

- A. Nothing
- ✓ **B. \$3M in excess reprocurement costs, if the government reprocured reasonably**

- C. \$13M
- D. \$10M

### WHY

Excess reprocurement costs are the defaulted contractor's liability, subject to reasonableness.

**What happens when a termination for default is found improper?**

A. The contract is reinstated

✓ **B. It converts to a termination for convenience and the contractor recovers under those rules**

C. The contractor is debarred

D. Nothing changes

**WHY**

Conversion is the remedy for defaults issued without proper process or in the face of excusable delay.

**Which of these is an excusable delay under the default clause?**

A. A subcontractor's delay that the prime could have avoided

✓ **B. Unusually severe weather beyond the contractor's control**

C. The contractor's own staffing shortfall

D. A supplier price increase

**WHY**

Excusable delays are beyond the contractor's control and without its fault; subcontractor delays qualify only if beyond both parties' control.

**Within what period must a termination settlement proposal be submitted?**

A. 30 days

B. 90 days

✓ **C. One year**

D. Three years

**WHY**

One year from the effective date of termination, on the standard forms.

**Why should a contractor open new charge numbers immediately after a convenience termination?**

- A. To increase fee
- ✓ **B. To segregate settlement expenses from the work so the proposal is auditable and settles quickly**
- C. The FAR requires it
- D. To bill the subcontractors

**WHY**

Mixed records turn a months-long settlement into years.

**What is the most important document a contractor writes when facing default?**

- A. The invoice
- ✓ **B. The response to the cure or show cause notice: facts, excusable causes, government dependencies, and a recovery plan with dates**
- C. The CPARS rebuttal
- D. The settlement proposal

**WHY**

That response is the record a board of contract appeals reads.

**How does a default affect future competitions?**

- A. It does not
- ✓ **B. It is reported in past performance systems, must be disclosed, and becomes a responsibility question on future awards**
- C. It improves the CPARS score
- D. It is confidential

**WHY**

For a small company, a single default can be fatal to future awards.

**Match the item to its treatment after a convenience termination:****CORRECT MATCHES**

|                                         |                                 |
|-----------------------------------------|---------------------------------|
| Costs incurred on terminated work       | Recoverable                     |
| Anticipated profit on unperformed work  | Not recoverable                 |
| Settlement expenses                     | Recoverable                     |
| Costs of continuing after the stop date | Not recoverable unless directed |

**WHY**

Incurred costs and settlement expenses are recoverable; profit on work performed is recoverable; anticipated profit on unperformed work is not; costs after the stop date are not.

**Order the government's process toward a default for failure to make progress:****CORRECT ORDER**

1. Apply the QASP remedies and document nonperformance
2. Issue a cure notice with a period to cure or explain
3. Receive and evaluate the contractor's response
4. CO assesses excusable delay and government contribution
5. Issue the notice of default, or withdraw

**WHY**

QASP remedies, cure notice, contractor response, CO decision considering excusable delay and government contribution, then notice of default.

**Why does an improper default cost the government twice?**

A. It does not

✓ **B. The default converts to convenience (a settlement) and the government still pays for the reprocurement itself**

C. Because of interest

D. Because of protests

**WHY**

Defaulting a contractor with a good defense produces a settlement and an unrecoverable reprocurement.

## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 6

# Contract Transition: Taking Over and Handing Off Without Dropping the Mission

24 min

*The 30 to 90 days when one contractor leaves and another arrives are the riskiest period in a service contract's life. What transition-in and transition-out actually involve, the incumbent's obligations and incentives, workforce capture, knowledge transfer, government-furnished property and access, the government's role, and the plan that gets a new contractor to full performance on day one.*

## The Mission Does Not Pause for the Handoff

When a service contract changes hands, the help desk still has to answer, the network still has to run, the aircraft still have to be maintained. The government is paying for continuous performance and the two contractors are, at that moment, competitors: one leaving, often unwillingly, and one arriving without the people, the access, or the knowledge it needs. Transitions fail more often than they should, and the failure is visible: dropped tickets, missed maintenance, systems down, a COR fielding complaints for two months. This lesson covers what a transition involves from both sides and from the government's seat, and the plan that gets a new contractor to full performance on day one.

### IN PLAIN TERMS

Think of a relay race. The baton is passed inside an exchange zone where both runners are moving at full speed, side by side, for a few strides. If the incoming runner starts too late, or the outgoing runner lets go early, the baton hits the track and the race is lost. A contract transition is that exchange zone: both contractors run in parallel for a planned period, and the government's job is to keep the baton from being dropped.

## The Three Parties and What Each Wants

| PARTY               | WANTS                                                                                                                      | OBLIGATIONS                                                                                                           | LEVERAGE                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Incoming contractor | Staff, access, knowledge, and time to reach day-one readiness; the incumbent's employees                                   | The transition plan it proposed; performance from the handoff date                                                    | The new contract; the government's interest in its success              |
| Outgoing contractor | To finish honorably, collect what it is owed, keep its people for other work, and sometimes to make the successor look bad | Continuity of services clause; transition-out requirements in its contract; property return; cooperation              | The knowledge and the people; the government's dependence until handoff |
| Government          | No degradation in performance; no gap; the incumbent's cooperation; a successor ready on day one                           | Provide access, accounts, badges, GFP, and decisions on time; enforce the incumbent's obligations; manage the overlap | Both contracts; the CPARS on each                                       |

## Transition-In: What the Incoming Contractor Has to Do

From award to full performance, typically 30 to 90 days, the incoming contractor must: recruit and hire the workforce (usually including a large share of the incumbent's staff), obtain clearances, badges, system accounts, and facility access for every one of them, take over government-furnished property and systems, receive and absorb the incumbent's documentation, procedures, and open work, establish its own management, reporting, and quality processes, and begin performing, first shadowing, then leading, then alone. Each of these has dependencies on the incumbent and on the government, and each is on the critical path. The transition plan sequences them; the transition manager, a named person whose only job is the transition, runs them daily.

## The Transition Plan

What it contains, whether it is a proposal volume or a post-award deliverable.

- **Milestones and schedule** Day by day for the first two weeks, weekly after; award, kickoff, staffing waves, access, knowledge transfer sessions, shadowing, assumption of each function, full performance.
- **Staffing plan** Positions, sources (incumbent capture, internal transfers, new hires), offer timing, clearance and badging status per person, contingency for declines.
- **Access and GFP** Every account, badge, clearance, facility, system, and item of property needed, who provides it, and the requested date.
- **Knowledge transfer** Documentation to receive, sessions to hold, systems to be walked through, open items and their status, and how tacit knowledge will be captured (shadowing, interviews).
- **Risk register** The transition-specific risks (key incumbent staff decline, access delayed, incumbent uncooperative) with mitigations.

- **Government dependencies** Everything the government must do and by when, stated explicitly, because government delays are the most common transition risk.
- **Communication** Daily standups with the COR during the critical period; weekly reports; a single point of contact.

## Incumbent Capture: The Center of Most Transitions

On a services contract, most of the knowledge is in the people, and the fastest transition is one where the people stay. The incoming contractor typically wants to hire most of the incumbent's staff; the incumbent may want to keep them for other work, or may have no other work and let them go. The rules: the incoming contractor may recruit them (non-solicitation clauses in the incumbent's employment agreements are sometimes asserted but rarely enforceable against a successor on a government contract); the incumbent must not obstruct; the government may include workforce continuity provisions in the solicitation where current policy allows (the executive order requiring right of first refusal has been imposed and rescinded; check the contract). The practical work is offer letters within days of award, competitive terms, a job fair at the site, and a clear message that the incumbent's people are wanted. Employees decide in the first week whether they trust the new company. The transition manager's first job is that week.

### EXAMPLE (ILLUSTRATIVE)

A program office awards a help desk contract to a new company with a 60-day transition. The incumbent has 40 people on site. The incoming transition manager arrives the day after award, holds a job fair in week one, and makes offers to all 40. 32 accept, which is  $32 \div 40 = 80$  percent of the workforce. The company recruits the other 8 from its own staff and new hires. The risk now sits with the government. Each new hire needs a badge and system accounts. If the government delivers accounts in week one, the new staff shadow the incumbent for seven weeks. If accounts arrive in week six, five of those weeks are lost and cannot be recovered, which is why the plan lists government dependencies with dates.

### ONLY THE GOVERNMENT CAN ENFORCE TRANSITION-OUT

The incumbent who lost the recompetes has every incentive to make the transition hard and a contractual obligation to make it smooth. Most behave professionally; some do not. The government is the only party that can enforce the incumbent's transition-out obligations, and it should, early and in writing: the continuity of services clause, the transition-out requirements in the PWS, and the CPARS that is still being written. An incoming contractor that cannot get the incumbent's cooperation goes to the COR and the CO, not to the incumbent's program manager.

## The Government's Role: The Party That Controls the Critical Path

Government actions are on the critical path of every transition: system accounts, badges, facility access, clearance transfers, GFP transfer, and the decisions that only the government can make. A transition in

which the government provides accounts in week six instead of week one has lost five weeks that no contractor can recover. The government PM and COR should treat their own transition tasks as deliverables with dates, hold a kickoff with both contractors, direct the incumbent's cooperation in writing, receive the incoming contractor's dependency list and commit to it, and be available daily. The CPARS for both contractors covers the transition, and the government's own performance during it is what the contractors will remember.

One sentence to teach your team: **a transition succeeds when the incoming contractor has a named manager running a dated plan, captures the incumbent's people in the first week, and gets the government to deliver access and property on its own critical path, while the government enforces the incumbent's obligation to cooperate.**

#### INTERMEDIATE

### Transition-Out: The Incumbent's Obligations and How to Meet Them Well

The outgoing contractor's contract usually includes transition-out requirements: a transition-out plan, knowledge transfer, documentation delivery, cooperation with the successor, property return, and continued performance to the handoff date. The continuity of services clause adds that the contractor must continue performing and cooperate for a period after the contract ends if the government asks, and be paid for it. Meeting these well is not only an obligation; it is the incumbent's last CPARS, its reputation with a customer it will want again, and often the difference between an orderly departure and a dispute. The professional practice: a transition-out manager, a documentation package delivered early, structured handoff sessions, honest status on open items, retention bonuses to keep key staff through the handoff, and a final report. Withholding cooperation to punish the customer for the recompetete decision costs the incumbent more than it costs anyone else.

## Transition Timeline (90-Day Model)

| PERIOD         | INCOMING CONTRACTOR                                                                               | OUTGOING CONTRACTOR                                                           | GOVERNMENT                                                                           |
|----------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Award to day 7 | Transition manager on site; kickoff; offers to incumbent staff; access requests submitted         | Notified; transition-out plan activated; staff informed                       | Kickoff with both; direct cooperation in writing; start account and badge processing |
| Days 8 to 30   | Staffing waves; clearances and badges; receive documentation; begin shadowing                     | Deliver documentation; knowledge transfer sessions; continue full performance | Deliver accounts and access; transfer GFP inventories; daily standups                |
| Days 31 to 60  | Shadow all functions; assume low-risk functions; management processes stood up                    | Support shadowing; hand off functions as directed; retain key staff           | Direct function-by-function handoff; monitor performance; resolve disputes           |
| Days 61 to 90  | Assume remaining functions; incumbent shadows the successor; full performance readiness confirmed | Phase down; final property return; final report                               | Confirm day-one readiness; accept transition deliverables; close out the incumbent   |
| Day 91 onward  | Full performance; QASP applies                                                                    | Continuity of services if invoked; closeout                                   | Surveillance; CPARS on both                                                          |

## Knowledge Transfer That Actually Transfers

Documentation is necessary and insufficient. The knowledge that matters most is tacit: which customer wants what, which system fails how, which workaround keeps the process running, which open item is about to become a problem. It transfers through people (incumbent capture), through shadowing (the incoming staff watching and then doing, with the incumbent watching), and through structured sessions where the incumbent walks through each function, each system, and each open item with the successor and the government present. The incoming contractor should arrive with a list of questions per function and leave each session with answers recorded. The government should attend, because the government is the party that will still be there and because its presence keeps the session honest.

## Transition in the Proposal and the Evaluation

Transition approach is usually an evaluation factor on a services recompetes, and it deserves the attention capture teams give the technical approach. Evaluators look for a realistic schedule with government dependencies stated, a credible staffing plan with a specific incumbent capture approach, a named transition manager with relevant experience, and identified risks with mitigations. The incumbent proposes "no transition risk," which is true and which evaluators discount; a challenger proposes a plan detailed enough to be believed. On the government side, the solicitation sets the transition period length, whether the overlap is funded, and what the incumbent's transition-out obligations are, and those choices determine how feasible a change of contractor is. A solicitation with a two-week transition and no funded overlap has effectively chosen the incumbent.

**ADVANCED**

## Transition Economics

Transition costs money on all three sides. The incoming contractor spends on recruiting, retention bonuses, a transition team, and an overlap period during which it is staffed but not yet fully billing; those costs are priced into the transition CLIN if the solicitation has one, or into the first year if it does not. The outgoing contractor spends on retention and cooperation with declining revenue; the continuity of services clause pays for extended performance but not for goodwill. The government funds the overlap and pays twice for the same period. Each party economizes in ways that raise the others' risk: an incoming contractor that skips retention bonuses loses the incumbent's staff; an incumbent that releases staff early leaves nothing to transfer; a government that shortens the overlap saves money and buys a degraded first quarter. Senior PMs on all sides know the transition is cheaper than the failure and budget accordingly.

## When Transitions Go Wrong

The failure patterns are consistent. The incoming contractor's offers arrive late or below the incumbent's pay, and the staff leave for other work. Government accounts and badges take weeks and the new staff sit idle. The incumbent delivers documentation that is out of date and sends its best people elsewhere. Open items are not disclosed and become the successor's crises. Government-furnished property is not inventoried and disputes about its condition consume the first month. The successor assumes full performance before it is ready because the calendar said so. Each has a remedy in the plan: early offers at competitive terms, government dependencies with dates and escalation, documented and verified knowledge transfer with the government present, a joint property inventory, and a readiness review before assumption of each function with the authority to delay it.

## The Government PM's Transition Checklist

What the government owns and when.

**Before award: design the transition** **SOLICITATION**

Transition period length, funded overlap, transition-out obligations in the incumbent's contract, transition as an evaluation factor.

A solicitation without these has chosen the incumbent by default.

**Award week: kickoff and direction** **START**

Joint kickoff; written direction to the incumbent on cooperation; account and badge requests started the same day.

The government's first-week actions set the pace for everything.

**Weeks 2 to 8: deliver and enforce** EXECUTE

Access and property delivered on the committed dates; incumbent cooperation enforced; daily standups.

Government delays are the most common transition risk. Track your own tasks like deliverables.

**Function handoffs: readiness reviews** HANDOFF

Each function transfers after a review confirms the successor is ready, not on a calendar date.

The authority to delay a handoff is what prevents a premature assumption.

**After: CPARS on both and lessons learned** CLOSE

The transition is part of both contractors' performance record; the government's own lessons go into the next solicitation.

The next recompetes transition design starts with this one's lessons.

**WHAT SENIOR PMS DO DIFFERENTLY**

Incoming: they put a full-time transition manager on site the day after award, make offers to the incumbent's staff in the first week, and state every government dependency with a date. Outgoing: they transition out professionally because the CPARS and the customer relationship outlast the contract.

Government: they design the transition in the solicitation, deliver their own critical-path items on time, enforce the incumbent's cooperation in writing, and hand off functions on readiness rather than on the calendar.

**BUILD ON THIS**

- **Common Mistakes Contractors Make** Re compete planning starts at award
- **Services Acquisition** Designing services for competition and continuity
- **Contract Administration & COR** The COR's role during transition

## Key Terms

### Transition-In

The period, typically 30 to 90 days, during which the incoming contractor stands up: staffing, access, knowledge transfer, and assumption of work, often overlapping the incumbent's final weeks.

### Transition-Out

The outgoing contractor's obligations at the end of its contract: cooperating with the successor, transferring knowledge and materials, returning property, and continuing performance until the handoff date.

### Transition Plan

The document, usually a proposal deliverable and then a contract deliverable, laying out the incoming contractor's activities, milestones, staffing, risks, and dependencies for the transition period.

### Incumbent Capture

Hiring the outgoing contractor's employees to continue in their roles under the new contract. The fastest path to continuity and the most contested part of a transition.

### Knowledge Transfer

The structured handoff of documentation, procedures, system access, open items, and tacit knowledge from the outgoing to the incoming contractor.

### Government-Furnished Property and Information (GFP/GFI)

Equipment, facilities, systems, data, and access the government provides. Transition depends on their timely transfer from incumbent to successor and on the government's own actions (accounts, badges, clearances).

### Phase-In / Phase-Out

The overlapping period in which both contractors are on contract: the incumbent phasing out, the successor phasing in. Its length and funding are set by the solicitation.

### Transition Risk

The risk that mission performance degrades during the handoff: lost staff, lost knowledge, access gaps, unfinished work. Evaluated in source selection and managed through the transition plan.

### Continuity of Services Clause

The clause (FAR 52.237-3) requiring the outgoing contractor to continue services and cooperate in the transition for a set period after contract end, if the government requests it.

### Day-One Readiness

The state in which the incoming contractor can perform the full scope at the required standard from the first day of full performance: staffed, cleared, badged, trained, with access and tools.

## Knowledge Check

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**What is the most common transition risk?**

- A. Contractor fraud
- ✓ **B. Government delays in providing accounts, badges, access, and property**
- C. Weather
- D. Protests

### WHY

Government actions are on the critical path of every transition and are the ones no contractor can recover.

**What does the continuity of services clause require of the outgoing contractor?**

- A. Nothing after contract end
- ✓ **B. Continued performance and cooperation in the transition for a period after contract end if the government requests, with payment**
- C. Free work for 90 days
- D. Transfer of its employees

### WHY

FAR 52.237-3 gives the government a paid extension of the incumbent's services to prevent a gap.

**Why is incumbent capture central to most services transitions?**

- A. Because it is required
- ✓ **B. Because most of the knowledge is in the people, and the fastest transition is one where the people stay**
- C. Because it lowers price
- D. Because the government mandates it

### WHY

Offers in the first week at competitive terms decide whether the incumbent's staff stay.

**Who can enforce the incumbent's transition-out obligations?**

A. The incoming contractor

✓ **B. The government, through the CO and COR, in writing, with the CPARS still open**

C. The incumbent's employees

D. No one

**WHY**

The incoming contractor escalates to the COR and CO, not to the incumbent's PM.

**What should a transition plan state explicitly that most omit?**

A. The price

✓ **B. Every government dependency with a requested date**

C. The incumbent's salaries

D. The CPARS score

**WHY**

Stating government dependencies makes them trackable and creates the record when they slip.

**When should a function transfer from the incumbent to the successor?**

A. On the calendar date in the plan

✓ **B. After a readiness review confirms the successor is ready, with authority to delay**

C. When the incumbent decides

D. On day one of the new contract

**WHY**

Premature assumption because the calendar said so is a recurring failure.

**A solicitation provides a two-week transition with no funded overlap. What has the government effectively done?**

A. Saved money with no consequence

✓ **B. Chosen the incumbent by default, because a change of contractor is infeasible on those terms**

C. Complied with the FAR

D. Increased competition

**WHY**

Transition design in the solicitation determines whether a challenger can realistically win and perform.

**How does tacit knowledge transfer?**

A. Through documentation only

✓ **B. Through people (incumbent capture), shadowing, and structured sessions per function with the government present**

C. Through the contract file

D. It does not

**WHY**

Documentation is necessary and insufficient. The government's presence keeps sessions honest.

**Why should the outgoing contractor transition out professionally even after losing the recompet?**

A. It is not required

✓ **B. Because the CPARS is still being written, the customer relationship outlasts the contract, and obstruction costs the incumbent most**

C. To receive a bonus

D. To avoid a protest

**WHY**

Withholding cooperation to punish the customer is a reputational and contractual loss.

**Match the party to what it controls:****CORRECT MATCHES**

|                     |                                                         |
|---------------------|---------------------------------------------------------|
| Incoming contractor | The transition plan, the offers, the transition manager |
| Outgoing contractor | The knowledge, the people, cooperation                  |
| Government          | Accounts, badges, GFP, and enforcement of obligations   |

**WHY**

Incoming controls the plan and offers; outgoing controls knowledge and people; government controls access, property, and enforcement.

**Order a 90-day transition:****CORRECT ORDER**

1. Award week: transition manager on site, kickoff, offers to incumbent staff, access requests
2. Days 8 to 30: staffing waves, badges and accounts, documentation, shadowing begins
3. Days 31 to 60: shadow all functions, assume low-risk functions
4. Days 61 to 90: assume remaining functions after readiness reviews
5. Day 91: full performance under the QASP

**WHY**

Kickoff and offers, staffing and access, shadowing and function handoffs, full performance readiness.

**Which economizing choice raises another party's risk?**

- A. The incoming contractor paying retention bonuses
- ✓ B. The government shortening the funded overlap to save money
- C. The incumbent delivering documentation early
- D. The government attending knowledge transfer sessions

**WHY**

Each party can save money in ways that degrade the transition; the transition is cheaper than the failure.

## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 7

# Options, Extensions, and Bridges: Keeping a Contract Alive Legally

22 min

*How a contract continues past its base period. Option exercise and the rules that make it valid (notice, timing, the determination, funds), the six-month extension clause, what happens when an option is missed, bridge contracts and why they are scrutinized, and how the option decision is really made on both sides.*

## The Contract Continues Only If Someone Acts

Most service contracts are a base year and four option years. The options are not automatic. Each one is a decision the government makes, documented and exercised in a specific way within a specific window, and if the process is not followed, the contract ends whether anyone intended it to or not. Contracting offices have let contracts lapse by missing a notice date; contractors have found themselves performing without a contract; programs have paid for bridges because the recompetes were not ready. This lesson covers how options are exercised properly, what happens when they are not, the six-month extension that exists for emergencies, and the bridge contract that exists for planning failures.

## How an Option Is Exercised Correctly

The option clause sets the mechanics. The government gives preliminary notice of its intent to exercise, in writing, a set number of days before the current period ends (often 60 days). Then it exercises, by a unilateral modification, before the current period expires (the clause states the deadline). Before exercising, the CO makes the FAR 17.207 determination: funds are available; the requirement still exists and the option fulfills it; exercising is the most advantageous method, considering price and other factors (the CO may need to check the market, informally, to confirm the option price is still fair); and the option was evaluated as part of the original award. The exercise must be exactly on the option's terms. An exercise that is late, that changes the price, or that adds conditions is not an exercise; it is an offer the contractor may accept or decline.

## The Option Exercise Checklist

- **Calendar the dates** The preliminary notice deadline and the exercise deadline, for every option on every contract, with reminders months ahead. Missing either is the most common failure.
- **Confirm funds** The option must be funded (or the exercise made subject to availability of funds, with the clause and the language that allow it) before exercise.
- **Confirm the requirement** Does the program still need this, at this quantity, for this period?

- **Confirm advantageousness** Is the option price still fair against the market? Has performance justified continuing?
- **Confirm it was evaluated at award** If the option was not part of the evaluated price, exercising it may be a new procurement requiring competition.
- **Document the determination** The FAR 17.207 finding, signed by the CO, in the file.
- **Issue the preliminary notice, then the exercise** In writing, on time, on the option's exact terms.

## When an Option Is Missed

If the government fails to give timely notice or exercise on time, the option lapses and the contract ends at the current period's expiration. The government cannot unilaterally revive it. What it can do is ask the contractor to agree, by bilateral modification, to continue on the option's terms or on renegotiated terms; the contractor is under no obligation to agree and can ask for a price adjustment, which is where the government pays for the missed date. If the contractor declines, the government needs a new contract, which means a J&A for a bridge or a gap in service. Contractors should watch the government's option calendar as closely as the government does, both to protect the revenue and to know when leverage shifts.

### EXAMPLE (ILLUSTRATIVE)

A program office has a services contract with a base year and four option years. The option clause requires preliminary notice 60 days before the current period ends and exercise before it expires. Option Year 2 is priced at \$4,200,000. The contract specialist leaves, nobody calendared the dates, and the exercise deadline passes. The option lapses and the contract ends at the end of the current period. The government cannot revive it on its own. The CO asks the contractor to continue by bilateral modification. The contractor agrees but asks for a 4 percent adjustment for updated labor costs:  $\$4,200,000 \times 0.04 = \$168,000$ , for a new price of \$4,368,000. The missed date cost the government \$168,000, and if the contractor had declined, a bridge with a J&A.

### THE OPTION IS NOT AUTOMATIC

An option year is the single largest booking most contractor PMs will produce in a year, and it depends on a government employee sending a letter on time. The contractor PM who confirms the option calendar with the CO six months out, keeps performance strong through the decision window, and has the CPARS record ready is the one whose options get exercised on time. The one who assumes the option is automatic finds out otherwise in the last week of the period.

## The Six-Month Extension: FAR 52.217-8

Most service contracts include the option to extend services clause, which lets the government require continued performance for up to six months, in increments, at the contract's existing rates. It exists to cover

a gap when the follow-on is delayed: the government can extend the incumbent for a month or three or six while the new award is completed. The clause requires the CO to give notice within the period it specifies before the contract expires. It is a genuine option, exercised unilaterally, and the contractor must perform. Two constraints: the six months is a total across all uses under that clause on that contract, and, because it was evaluated at award (or should have been; solicitations now typically state that the six months are included in the evaluated price), it does not require a new competition. Used for its purpose, the clause is the orderly answer to a late follow-on. Used repeatedly to avoid a recompetes, it is a bridge in disguise and draws the same scrutiny.

One sentence to teach your team: **options continue a contract only if the government gives notice and exercises on time and on the option's exact terms after a documented determination, a missed option can only be revived by the contractor's agreement, and the six-month extension clause covers a gap while a bridge contract covers a planning failure.**

#### INTERMEDIATE

### The Option Decision From the Government Side

The CO's determination is the formal step, but the decision is made in the program office. The questions: Is the contractor performing (CPARS, QASP results, award fee history)? Is the requirement unchanged, or has it grown or shrunk enough that the option no longer fits (in which case a modification or a new competition may be needed)? Is the price still fair, given market changes and the wage determination updates that the price adjustment clause will bring? Are the funds in place for the option period, in the right appropriation and year? Is there a reason to compete instead (a small business now capable, a policy push, a chance to restructure)? A government PM should answer these six months before the notice date, because a decision not to exercise means a recompetes that should already be underway.

## Option Mistakes and Their Consequences

| MISTAKE                                       | CONSEQUENCE                                                            | PREVENTION                                                                 |
|-----------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Missed preliminary notice                     | Exercise may be invalid; contractor can decline or renegotiate         | A calendar with reminders; the contract specialist and COR both tracking   |
| Exercise after the period expired             | Not an exercise; contract has ended; bridge or bilateral needed        | Exercise early in the window, not on the last day                          |
| Exercise with changed terms                   | Counteroffer, not exercise; contractor may accept, decline, or reprice | Exercise exactly; negotiate changes separately by bilateral mod            |
| Exercise without funds                        | Anti-Deficiency Act exposure unless subject to availability of funds   | Confirm funding; use the availability-of-funds clause only where permitted |
| Option not evaluated at award                 | Exercise may be an improper sole-source procurement                    | Include options in the evaluated price; document it                        |
| Serial 52.217-8 extensions to avoid recompile | Scrutiny; six-month cap reached; J&A required beyond it                | Start the recompile on time                                                |

## Bridge Contracts: What They Are and Why Everyone Watches Them

When the follow-on is not ready and the six months are used up (or the contract has no extension clause), the government awards a bridge: a sole-source contract or modification extending the incumbent for months, justified by a J&A under urgency or only-one-source. The J&A lesson covered why these are hard to write honestly. They are also tracked: inspectors general and Congress have reported repeatedly on bridge contracts as evidence of acquisition planning failure, some agencies require senior approval and a corrective action plan for each one, and a program with serial bridges is a program that has stopped competing. For the incumbent, a bridge is revenue with a warning: the government will eventually compete the work, and the bridge period is when the challengers are preparing.

## Options in the Proposal and the Price

Options are priced at award and the government evaluates the total including options, which is why option-year pricing is a capture decision. Pricing options low to win the evaluation and hoping to renegotiate later fails: the government exercises at the option price, and the price adjustment clause covers only wage determination increases, not the contractor's pricing regret. Pricing options high to protect margin loses the evaluation. The discipline is to price each option year at what it will actually cost, with escalation that reflects the company's merit history above the wage determination floor, and to remember that the government can decline any option and recompile. The option years are the contractor's to lose through price or performance.

## ADVANCED

## Options, Recompets, and the Two-Year Rule of Thumb

The option structure and the recompile timeline interact. If the government intends to recompile at the end of the options, the recompile must start about two years before the final option ends, which is during the second-to-last option year. If the government is undecided about exercising the next option, it must decide six months before the notice date so that a recompile, if chosen, can start. Senior government PMs therefore treat each option decision as a fork: exercise, or start the recompile now. Deciding late produces the worst outcome: an option exercised for lack of an alternative, followed by a rushed recompile, followed by a bridge. The pre-award timeline lesson applies to every option year, not just to the original award.

## Total Contract Length and Its Limits

Service contracts are generally limited to a total of five years including options (with statutory and agency exceptions for certain services and for multi-year contracts under their own rules), and vehicle ordering periods are limited by their own terms. A program that wants ten years of continuous service structures it as two contracts with a recompile between, or as a vehicle with a long ordering period and orders within it. Options cannot extend a contract beyond its stated total, and an extension beyond the total is a new contract requiring competition or a J&A. Government PMs planning a long-lived service should design the sequence of contracts and recompets across the decade rather than discovering the five-year limit in year four. Watch the FAR overhaul: a September 2026 proposed rule (FAR Case 2026-006) would replace the general five-year limit with whatever statute or regulation applies. Until it is final, plan to five.

## The Option Year Playbook, Both Sides

What happens in the twelve months before an option date.

### 12 months out: The fork DECIDE

Government: exercise, or start the recompile. Contractor: confirm the calendar with the CO; assess performance standing.

A recompile started now finishes in time. One started at six months does not.

### 9 months out: Requirement and price check VERIFY

Government: has the requirement changed; is the price fair; is the wage determination update coming. Contractor: prepare the price adjustment request; document performance.

A changed requirement may need a modification before or with the exercise.

**6 months out: Funding** MONEY

Government: funds identified for the option period, right appropriation and year. Contractor: forecast the option year; staffing plan ready.

Unfunded options are exercised subject to availability of funds only where the clause and policy allow.

**3 months out: Determination and notice** ACT

Government: FAR 17.207 determination signed; preliminary notice issued on time. Contractor: acknowledge; confirm readiness.

Issue the notice early in the window, not on the deadline.

**Before expiration: Exercise** EXECUTE

Government: unilateral modification on the option's exact terms, before the period ends. Contractor: confirm receipt; file the price adjustment request within 30 days of the new WD.

The exercise is the booking. The price adjustment request is the margin.

**WHAT SENIOR PMS DO DIFFERENTLY**

Government: they treat each option date as a fork between exercising and recompeting, decided a year out; they calendar the notices and exercise early in the window; they document the determination; and they use the six-month clause for its purpose and bridges almost never. Contractor: they confirm the option calendar with the CO, keep performance strong through the decision window, price options at real cost, and file the price adjustment inside the window every year.

**BUILD ON THIS**

- **Contracts vs. Task Orders** Bridge contracts and option exercise discipline
- **J&As, D&Fs, and the Approvals That Gate an Award** The bridge J&A and the option D&F
- **Labor Standards** The price adjustment clause at each option

## Key Terms

### Option

A unilateral right in a contract by which the government may elect to purchase additional quantities or extend the term, at prices or terms set at award. The contractor is bound; the government is not until it exercises.

### Preliminary Notice

The written notice, required by the option clause a set number of days before the option date, that the government intends to exercise the option. Not itself the exercise; a missed preliminary notice can invalidate the exercise.

### Exercise

The government's written, unilateral modification exercising the option, issued within the time and on the terms the clause specifies. Must be exact; a late or conditional exercise is not an exercise.

### FAR 17.207 Determination

The CO's written finding, before exercising, that funds are available, the requirement still exists, the option is the most advantageous method of fulfilling it (price and other factors), and the option was evaluated at award.

### Option to Extend Services (FAR 52.217-8)

The clause allowing the government to require continued performance for up to six months at the contract's rates, exercised in increments, used to cover a gap while a follow-on is awarded.

### Option to Extend the Term (FAR 52.217-9)

The standard clause for exercising option years, with the preliminary notice period and the exercise deadline.

### Bridge Contract

A non-competitive extension or short-term award to an incumbent to cover the gap between an expiring contract and a delayed follow-on. Requires a J&A; a recognized indicator of planning failure.

### Bilateral Modification

A modification signed by both parties. When an option cannot be exercised as written (late, or on changed terms), continuing performance requires the contractor's agreement, usually with a price adjustment.

### Total Contract Period

The base plus all options, generally limited to five years for services (with exceptions), which is why option years are structured as they are.

### Option Evaluation at Award

The requirement that options be evaluated in the original competition (total evaluated price including options) if they are to be exercised without a new competition.

## Knowledge Check

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Is an option year automatic?

A. Yes

✓ **B. No; the government must give preliminary notice and exercise on time, on the option's exact terms, after a documented determination**

C. Only for services

D. Only if the contractor requests it

### WHY

A missed notice or a late exercise ends the contract at the current period's expiration.

What must the FAR 17.207 determination find before an option is exercised?

A. That the contractor is small

✓ **B. That funds are available, the requirement exists, the option is the most advantageous method, and it was evaluated at award**

C. That DCAA has audited

D. That a J&A is approved

### WHY

The determination is the CO's documented basis for exercising.

The government exercises an option but changes the price in the modification. What is this?

A. A valid exercise

✓ **B. A counteroffer, not an exercise; the contractor may accept, decline, or reprice**

C. A bridge contract

D. A termination

### WHY

An exercise must be exactly on the option's terms. Changes require a bilateral modification.

**The government misses the preliminary notice date and the contract period is about to end. What can the government do?**

A. Exercise anyway; the notice is a formality

**✓ B. Ask the contractor to agree by bilateral modification to continue, possibly at a renegotiated price, or award a bridge with a J&A**

C. Nothing; the contract must end

D. Issue a stop-work order

#### WHY

The government cannot unilaterally revive a lapsed option. The contractor gains leverage.

**What does FAR 52.217-8 allow?**

A. Unlimited extensions

**✓ B. Requiring continued performance for up to six months total at the contract's rates, in increments, to cover a gap**

C. A new competition

D. A price increase

#### WHY

The option to extend services is the orderly answer to a delayed follow-on, within a six-month total.

**Why are bridge contracts scrutinized?**

A. They are illegal

**✓ B. They are non-competitive awards justified by J&As and recognized as evidence of acquisition planning failure**

C. They are too short

D. They require DCAA audits

#### WHY

Inspectors general and Congress track them; some agencies require senior approval and corrective action plans.

**Why is pricing option years low to win the evaluation a mistake?**

A. It is not

✓ **B. The government exercises at the option price, and the price adjustment clause covers only wage determination increases, not pricing regret**

C. Options are never exercised

D. Low prices are rejected

**WHY**

Price each option year at real cost with realistic escalation. The government can also decline and recompet.

**If the government intends to recompet at the end of the final option, when should the recompet start?**

A. At the end of the final option

✓ **B. About two years before the final option ends, during the second-to-last option year**

C. Six months before

D. After a bridge

**WHY**

The pre-award timeline applies to every option year, not only the original award.

**What is the general total length limit for service contracts including options?**

A. One year

✓ **B. Five years, with statutory and agency exceptions**

C. Ten years

D. No limit

**WHY**

Long-lived services are designed as a sequence of contracts with recompetes, or as vehicles with orders.

**Match the clause or term to its function:****CORRECT MATCHES**

|                          |                                            |
|--------------------------|--------------------------------------------|
| FAR 52.217-9             | Option to extend the term (option years)   |
| FAR 52.217-8             | Option to extend services up to six months |
| FAR 17.207 determination | The CO's findings before exercising        |
| Bridge contract          | Sole-source gap coverage requiring a J&A   |

**WHY**

52.217-9 exercises option years; 52.217-8 extends up to six months; 17.207 is the determination; a bridge covers a planning gap with a J&A.

**Order the government's option year actions:****CORRECT ORDER**

1. 12 months out: decide to exercise or start the recompet
2. 9 months out: verify the requirement and the price
3. 6 months out: identify funds for the option period
4. 3 months out: sign the determination and issue preliminary notice
5. Before expiration: exercise by unilateral modification on exact terms

**WHY**

Decide at the fork, verify requirement and price, confirm funding, sign the determination and give notice, then exercise before expiration.

**From the contractor's side, why confirm the option calendar with the CO months ahead?**

A. To negotiate a higher price

**✓ B. Because the option is the largest booking of the year and depends on the government acting on time; confirming the dates and performance standing protects it**

C. Because the FAR requires contractor notice

D. To trigger the six-month extension

**WHY**

Assuming the option is automatic is how contractors learn otherwise in the last week of the period.

## BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · LESSON 8

# Program Reviews and the Baseline: Reporting Up Without Getting Ambushed

23 min

*The review cadence a program lives inside: program management reviews, the acquisition program baseline, deviations and breaches, the reports that go to the milestone decision authority and to Congress, and the discipline of reporting bad news early. How a PM runs reviews that inform rather than perform, and what a contractor should expect to see on the government's side of the table.*

## The Review Is Not the Point; the Decision Is

A program lives inside a cadence of reviews: the contractor's monthly review with the program office, the program office's review with the program executive, the executive's review with the component, and, for major programs, the department's reviews and the reports to Congress. Each has a format, a date, and an audience with authority over the program's money and future. PMs spend a large share of their time preparing for them, and the preparation often becomes the job. This lesson is about running reviews that produce decisions rather than performances: the baseline the program is measured against, the deviation process when it will not be met, the reports that go up, and the single discipline that matters most, which is that leadership should never learn bad news at a review.

### IN PLAIN TERMS

Think of hiring a contractor to renovate your house with a budget target of \$50,000 and a hard ceiling of \$55,000. One contractor calls the day they find rotted beams behind a wall, explains the cost and offers options. Another says nothing and you learn about it from the final invoice. You will hire the first one again. The acquisition program baseline (APB) works the same way: an objective, a threshold, and a PM who reports the problem the day they see it coming.

## The Review Ladder

| REVIEW                                           | CADENCE                     | AUDIENCE                                  | CONTENT                                                                       | DECISIONS                                                        |
|--------------------------------------------------|-----------------------------|-------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------|
| Contractor PMR                                   | Monthly                     | Government PM and staff;<br>contractor PM | IPMR or cost report, schedule, technical, risks, deliverables, staffing       | Corrective actions; direction within scope; escalation items     |
| Program office PMR                               | Monthly or quarterly        | PEO or program executive                  | Status against the APB and the plan; risks; funding; contracting actions      | Resource decisions; risk acceptance; escalation to the component |
| Component review (service acquisition executive) | Quarterly or on event       | Component acquisition executive and staff | APB status; deviations; DAES inputs; major contracting actions                | Deviation approval; rebaselining; milestone scheduling           |
| DAES (OSD)                                       | Periodic for major programs | OSD acquisition staff                     | Assessment across areas with color ratings                                    | Department attention; potential program reviews                  |
| Milestone or decision review                     | At milestones and on breach | The milestone decision authority          | The full case: strategy, baseline, cost estimate, test results, risk, funding | Authority to proceed; conditions; the decision memorandum        |
| Congressional reporting                          | Annual (SAR); on breach     | Congress and the public                   | Cost, schedule, performance against baseline; breach explanations             | Funding; oversight; in critical breaches, program continuation   |

## The Acquisition Program Baseline: What the Program Promised

For major programs, the APB is the contract between the PM and the decision authority: for each key performance parameter, each schedule milestone, and each cost element (development, procurement, O&S, unit costs), an objective the program is aiming for and a threshold it must not breach. It is approved at Milestone B (and updated at C), and it is the standard for every deviation and every report. Programs without a formal APB (services, smaller efforts) have an equivalent: the approved cost, schedule, and performance plan they are measured against. The point is the same: the baseline is what the program said it would do, and reviews are about whether it is doing it. A PM who does not know their thresholds in dollar and date terms, the Common Mistakes lesson noted, cannot manage to them.

## Deviations: The Process When the Baseline Will Not Hold

When the PM determines that a threshold will not be met, a program deviation report goes to the MDA promptly (the policy sets a short period), stating the parameter, the cause, the magnitude, and the plan: recover, rebaseline, or accept. The MDA decides. For cost, the deviation process escalates to Nunn-McCurdy when unit cost growth crosses the statutory thresholds: significant breaches require congressional notification; critical breaches require the department to certify that the program is essential, that no cheaper

alternative exists, that the new estimates are reasonable, and that management is adequate, or terminate it. The discipline behind all of this is early identification: the deviation report is due when the PM determines the threshold will be breached, not when it is breached, and a PM who reports six months early gives the MDA six months of options.

#### EXAMPLE (ILLUSTRATIVE)

Program X's acquisition program baseline sets an average unit cost objective of \$20 million and a threshold of \$22 million (hypothetical figures). After a test finds a redesign is needed, the program's estimate climbs to \$21.5 million, still under the threshold. The cost team's trend analysis shows the redesign will push it to about \$22.6 million. That is \$22.6M minus \$20M = \$2.6M over objective, or 13 percent, and \$0.6M past the threshold. The breach has not happened yet, but the PM has now determined it will. So the PM calls the program executive officer that week, then sends the deviation report to the milestone decision authority with the cause, the magnitude and a plan. The next review discusses options instead of revealing a surprise.

#### THE REVIEW IS NEVER WHERE THE PROBLEM IS REVEALED

Leadership can absorb almost any bad news if they hear it first from the PM, early, with the cause and a plan. What they cannot absorb is hearing it at a review from a chart, or from someone else, or after the decision window has closed. The PMs who survive breaches are the ones who walked into the PEO's office the week they saw it coming. The review is where the plan is presented; it is never where the problem is revealed.

## Running a Review That Informs

- **Pre-brief the decision-maker** Anything significant is discussed with the principal before the review. The review confirms; it does not surprise.
- **Lead with status against the baseline** Cost, schedule, performance, risk against the thresholds, in the same format every time. Exceptions first.
- **One slide per problem: cause, impact, plan, ask** What happened, what it does to the baseline, what the program is doing, what it needs from the audience.
- **Decisions needed, listed** The review should end with recorded decisions and owners, not with a thank-you.
- **The forward look** What will change by the next review: the test, the option, the funding action, the risk that may realize.
- **Consistency with everything else** The numbers match the IPMR, the cost report, the MSR, and the dashboard. Inconsistency across products is what destroys credibility.

One sentence to teach your team: **reviews measure the program against the baseline it promised, a threshold that will not hold triggers a deviation report the moment the PM knows, and the discipline that keeps a PM employed is that leadership hears bad news from the PM first, early, with a plan.**

#### INTERMEDIATE

## The Contractor at the Government's Review

The contractor PM usually attends the contractor PMR and sometimes the program office review. What the contractor should understand is what the government PM is being measured on above that: the APB thresholds, the funding profile, the milestone schedule, the DAES colors. A contractor who knows that the government PM's schedule threshold is at risk understands why the government is pushing on a milestone that seems minor, and can help. A contractor who presents a rosy status at the contractor PMR while the government PM knows the IPMR says otherwise has lost credibility with the one person whose reporting decides the program's future. The Monthly Status Report lesson's advice applies: transparency and control matter more than perfection, because the government PM has to report up.

## What Each Level Wants to Know

| LEVEL                               | THE QUESTION                                                                 | WHAT ANSWERS IT                                                                                      |
|-------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Government PM (from the contractor) | Is the contract on plan, and if not, do you know why and what you are doing? | IPMR or cost report with honest variance explanations; risks with owners; a recovery plan with dates |
| PEO (from the PM)                   | Will the program meet its baseline, and what do you need from me?            | Status against thresholds; deviations early; specific asks: money, decisions, top cover              |
| Component acquisition executive     | Which of my programs need attention, and is this one being managed?          | Consistent reporting; deviations handled by process; a PM who is not surprised                       |
| OSD (DAES)                          | Is this program healthy across all assessment areas?                         | The data the program submits; the trend; the PEO's and component's assessment                        |
| Congress (SAR)                      | Is the program on cost and schedule, and if not, why?                        | The report, the breach explanation, the certification if critical                                    |

## Rebaselining: When and How

When a program cannot recover to its baseline, the alternative is a new one: a revised APB approved by the MDA with new thresholds reflecting the program's actual situation. Rebaselining is legitimate when the requirement, the funding, or the technical reality has changed and the old baseline no longer describes a

program anyone is executing. It is abused when it is used to erase variances without changing anything, which is the program-level version of the EVM rubber baseline. The MDA approves a rebaseline on evidence: a new independent cost estimate, a realistic schedule, a test program that supports the performance thresholds, and a funding profile that matches. Programs that rebaseline repeatedly are programs that have stopped meeting any baseline, and Congress tracks the number of times a major program's baseline has changed.

## The Decision Memorandum: Read the Conditions

A milestone or deviation review ends with a decision memorandum: the MDA's written decision, the direction, and the conditions. The conditions are the part programs forget. "Proceed to production, subject to completion of X by date Y and a report on Z" means the program has a new deliverable with a deadline and an audience. Senior PMs extract every condition from every decision memorandum into a tracker with owners and dates, because the next review will open with the conditions from the last one, and a program that has not met them has spent its credibility before the first slide.

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### ADVANCED

## Information Flow: Building the Reporting Stack Once

A program reports the same facts in six formats: the dashboard, the MSR, the IPMR, the PMR deck, the DAES input, the SAR. Each takes staff time, and each is an opportunity for inconsistency. Mature programs build one data stack (the Dashboard lesson's sources of record) and generate every format from it, so that the number on the PMR slide is the number in the IPMR is the number in the DAES. The staff time goes into analysis rather than assembly. The credibility comes from consistency. And when a reviewer at any level questions a number, the lineage from the format back to the source is the same for all of them. Programs that build each product separately spend their lives reconciling and their reviews defending.

## Reporting Bad News: The Practice

The theory is that bad news goes up early. The practice is a sequence. The PM sees the indicator (a leading indicator on the dashboard, a risk trend, an EAC movement, a test result). The PM verifies it within days, not weeks. The PM tells the PEO in person or by phone, before any document, with what is known, what is not, and when more will be known. The PM follows with the written deviation report or the memo, with cause, impact, and plan. The PM presents the plan at the next review, where the audience already knows the problem. And the PM reports progress against the plan at every review after, until it is closed. The sequence protects the program because it gives the decision authority time and options. It protects the PM because it is the record of a manager who saw the problem and acted. The alternative, hoping the problem resolves before the next review, is the ambush, and it ends careers.

## Review Anti-Patterns

What experienced reviewers recognize and discount.

### The green wall STATUS

Every chart green; the review ends early; the program breaches next quarter.

Reviewers learn that green means "not measured." Thresholds set by rule and applied honestly are the cure.

### The data dump VOLUME

Sixty slides, no exceptions highlighted, the problem on slide 47.

Exceptions first, one slide per problem, decisions listed. The appendix exists for the rest.

### The surprise AMBUSH

Leadership learns of a breach from the chart.

Pre-brief. Always.

### The inconsistent numbers CREDIBILITY

The PMR says one EAC; the IPMR says another; the DAES input a third.

One data stack, every format generated from it.

### The forgotten conditions FOLLOW-THROUGH

The last decision memorandum's conditions are not addressed.

A conditions tracker, opened before the next review.

## WHAT SENIOR PMS DO DIFFERENTLY

They know their thresholds in dollars and dates. They pre-brief every significant item. They report deviations the week they see them, with a plan. They build one data stack and generate every report from it. They track every condition from every decision memorandum. And they run reviews that end with recorded decisions, because a review that produces no decision was a performance.

## BUILD ON THIS

- **ACAT Levels** Nunn-McCurdy and the statutory reporting requirements
- **Stakeholder Management & Communication** Briefing flag officers and the 1-3-5 rule
- **Building a One-Page Program Dashboard** The data stack every report is generated from

## Key Terms

### Program Management Review (PMR)

The recurring review (monthly or quarterly) at which the program presents cost, schedule, performance, and risk status to the program executive or the customer. The working-level review that feeds everything above it.

### Acquisition Program Baseline (APB)

The formal agreement between the PM and the milestone decision authority on the program's cost, schedule, and performance parameters, each with an objective and a threshold. Approved at milestones; the standard against which deviations are measured.

### Objective and Threshold

For each APB parameter: the objective is the desired value; the threshold is the minimum acceptable (or maximum for cost and schedule). Exceeding a threshold is a deviation.

### Program Deviation

The PM's determination that the program will not meet an APB threshold. Requires a program deviation report to the MDA within a set period, with the cause and the plan.

### Nunn-McCurdy Breach

Statutory unit cost growth thresholds for major programs (significant and critical), measured against the current and original baselines, that trigger congressional notification and, for critical breaches, a certification or termination decision. Covered in the Foundations module; here it is the top of the deviation ladder.

### Selected Acquisition Report (SAR)

The annual (and on-breach) report to Congress on each major program's cost, schedule, and performance against the baseline. The public record of the program's health.

### Defense Acquisition Executive Summary (DAES)

The periodic internal OSD review of major programs across cost, schedule, performance, funding, contracting, and other assessment areas, with color ratings.

### Decision Memorandum

The written record of a milestone or review decision by the MDA, with the conditions and direction. The document that authorizes the program to proceed and constrains how.

### Reporting Up

The flow of program status from the PM to the PEO, the component acquisition executive, OSD, and Congress, each level with its own format and cadence.

### Ambush

Informal term for a review at which leadership learns bad news for the first time. The outcome every review discipline exists to prevent.

## Knowledge Check

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What is the acquisition program baseline?

A. The contractor's proposal

✓ **B. The formal agreement between the PM and the MDA on cost, schedule, and performance parameters, each with an objective and a threshold**

C. The IGCE

D. The EVM baseline

### WHY

The APB is the standard for deviations and reports. Programs without a formal APB have an equivalent approved plan.

When is a program deviation report due?

A. After the threshold is breached

✓ **B. Promptly after the PM determines the threshold will not be met**

C. At the next milestone

D. Annually

### WHY

Early identification gives the MDA options. The report is due on the determination, not on the breach.

What does a critical Nunn-McCurdy breach require?

A. A new contractor

✓ **B. The department to certify the program is essential, no cheaper alternative exists, the estimates are reasonable, and management is adequate, or terminate it**

C. A DCAA audit

D. A rebaseline only

### WHY

Critical breaches trigger the certification-or-termination decision and congressional notification.

**What is the single most important review discipline?**

A. Slide design

✓ **B. Leadership never learns bad news at a review; the PM pre-briefs early with cause and plan**

C. Keeping reviews short

D. Reporting only green

**WHY**

The review is where the plan is presented, never where the problem is revealed.

**When is rebaselining legitimate?**

A. Whenever variances are large

✓ **B. When the requirement, funding, or technical reality has changed and the old baseline no longer describes the program, supported by a new cost estimate, realistic schedule, and funding**

C. Never

D. Every year

**WHY**

Rebaselining to erase variances without changing anything is the program-level rubber baseline.

**What should a PM do with the conditions in a decision memorandum?**

A. File them

✓ **B. Extract them into a tracker with owners and dates, because the next review opens with them**

C. Ignore them until the next milestone

D. Delegate them to the contractor

**WHY**

Unmet conditions spend the program's credibility before the first slide.

**Why should a contractor understand the government PM's APB thresholds?**

A. To renegotiate price

✓ **B. Because the government PM is measured on them above the contractor PMR, and a contractor who understands them can help rather than surprise**

C. Because they are in the contract

D. To file a protest

**WHY**

Presenting rosy status when the IPMR says otherwise loses credibility with the person whose reporting decides the program's future.

**What is the Selected Acquisition Report?**

A. An internal OSD review

✓ **B. The annual and on-breach report to Congress on each major program's cost, schedule, and performance against baseline**

C. The contractor's status report

D. A GAO audit

**WHY**

The SAR is the public record of the program's health.

**Why build one data stack for all reporting formats?**

A. To reduce slides

✓ **B. So every format shows the same numbers with the same lineage, staff time goes to analysis, and credibility comes from consistency**

C. Because OSD requires it

D. To eliminate the IPMR

**WHY**

Programs that build each product separately spend their lives reconciling.

**Match the review to its audience:****CORRECT MATCHES**

|                             |                           |
|-----------------------------|---------------------------|
| Contractor PMR              | Government PM and staff   |
| Program office PMR          | Program executive officer |
| DAES                        | OSD acquisition staff     |
| Selected Acquisition Report | Congress                  |

**WHY**

Contractor PMR: government PM; program office PMR: PEO; DAES: OSD; SAR: Congress.

**Order the practice of reporting bad news:****CORRECT ORDER**

1. See the indicator on the dashboard or in a result
2. Verify within days
3. Tell the PEO in person or by phone before any document
4. Follow with the written deviation report: cause, impact, plan
5. Present the plan at the next review
6. Report progress against the plan until closed

**WHY**

See the indicator, verify quickly, tell the PEO in person, follow with the written report, present the plan at the review, report progress until closed.

**A review shows sixty slides, all green, with the problem on slide 47 and no decisions recorded. What is it?**

A. A thorough review

✓ **B. A performance, not a review: the green wall, the data dump, and no decisions**

C. A DAES input

D. A milestone review

**WHY**

Exceptions first, one slide per problem, decisions listed, and thresholds applied honestly.

**BEYOND AWARD: SUSTAINMENT, TEST, SOFTWARE, AND CLOSEOUT · MODULE ASSESSMENT**

# Module Assessment

Pass this to unlock the next skill level in the app.

## Knowledge Check

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**Roughly what share of a major weapon system's total lifecycle cost is typically operating and support?**

- A. About 10 percent
- B. About 30 percent

**✓ C. Well over half, often 60 to 70 percent**

- D. It is not tracked

### WHY

Development and production are the visible part. O&S over decades of service is the larger bill, and it is mostly locked in by early design decisions.

**Why does the government care about a contractor's data rights position when planning sustainment?**

- A. It affects the invoice format

**✓ B. Without technical data rights, only the OEM can compete depot and spares work, so sustainment becomes sole source for decades**

- C. Data rights only matter in research contracts
- D. It does not affect sustainment

### WHY

Data rights decided at development set who can compete sustainment. That is why they are a lifecycle cost decision, not a legal footnote.

**What is the basic difference between developmental test and operational test?**

A. DT is cheaper

✓ **B. DT checks whether the system meets its specification; OT checks whether typical users can accomplish the mission with it in realistic conditions**

C. OT is done by the contractor

D. There is no difference

**WHY**

A system can pass every DT specification and still fail OT because real operators in real conditions cannot use it effectively.

**Under the software acquisition pathway, what replaces the traditional milestone decisions?**

A. Nothing; milestones still apply

✓ **B. Frequent, iterative deliveries of working software with a minimum viable capability release and value assessments**

C. A single final acceptance test

D. A fixed-price production contract

**WHY**

The pathway is built around delivering usable capability early and often, with the program judged on value delivered rather than documents produced.

**On an agile contract, what is the most useful way to structure what the government is buying?**

A. A fixed list of features priced individually

✓ **B. Team capacity over time, with the backlog prioritized by the government product owner**

C. A single deliverable at the end

D. Whatever the contractor proposes

**WHY**

Buying capacity with government-controlled priorities lets the requirement evolve without a modification for every change.

**Which of these is required before a cost-type contract can be closed out?**

A. A press release

**✓ B. Final indirect rates settled and a final voucher submitted and accepted**

C. A new option exercised

D. A protest decision

**WHY**

Cost-type closeout waits on final rates, which is why it can trail physical completion by years.

**Why is closeout worth a contractor's attention even though the work is done?**

A. It is not; it is purely administrative

**✓ B. Unbilled costs, retained fee, and released funds sit unrecovered until closeout finishes, and records must be retained until it does**

C. It affects small business status

D. It sets the next contract's ceiling

**WHY**

Slow closeout is slow cash and long record-retention obligations. Both are real money.

**In a termination for convenience, what is the contractor generally entitled to recover?**

A. Nothing

B. Full anticipated profit on the whole contract

**✓ C. Costs incurred on the terminated work, reasonable settlement costs, and a fair profit on work performed, but not anticipatory profit**

D. Only costs already invoiced

**WHY**

T4C makes the contractor whole for work done and winding down, not for the profit it hoped to earn on work never performed.

**A contractor receives a cure notice. What is the right response?**

A. Ignore it; cure notices are routine

**✓ B. Respond in writing within the stated period with a specific, credible plan and evidence that the failure is being fixed**

C. Stop work immediately

D. File a protest

**WHY**

The cure notice is the last step before a termination for default. The written response is the record that a board or court will read later.

**During an incumbent-to-successor transition, what most often causes the service gap?**

A. The successor's pricing

**✓ B. Personnel, badges, system access, and government-furnished property not being ready on day one**

C. The contract number changing

D. Weather

**WHY**

Transition failures are almost always logistics: people not cleared, accounts not created, property not transferred.

**What does an option exercise require that a bridge does not?**

A. A J&A

**✓ B. That the option was evaluated at award, the government gives timely notice, funds are available, and the CO determines exercise is the most advantageous method**

C. New competition

D. Nothing; they are identical

**WHY**

Options are pre-priced and pre-competed. Bridges are sole-source extensions beyond the contract, which is why they need a justification and draw scrutiny.

**What does an acquisition program baseline breach signal?**

A. The program is cancelled automatically

**✓ B. Cost, schedule, or performance has moved past the approved threshold and must be reported and rebaselined with leadership review**

C. The contractor is in default

D. Funding has been increased

**WHY**

A breach is a reporting trigger, not an automatic penalty. The point is forcing the honest conversation before the gap grows.